# Radxa ROCK 5T

Schematic

V1.2

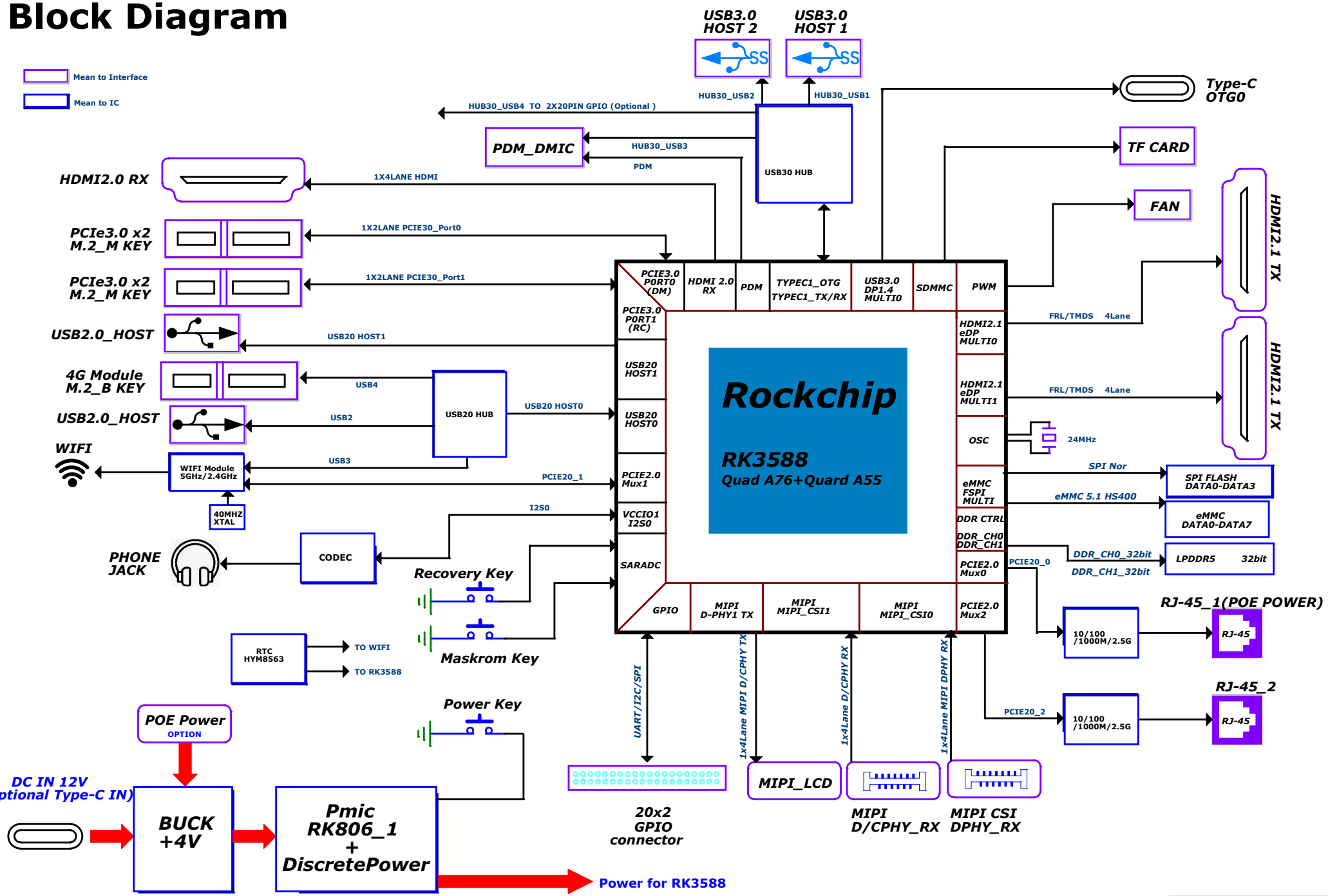
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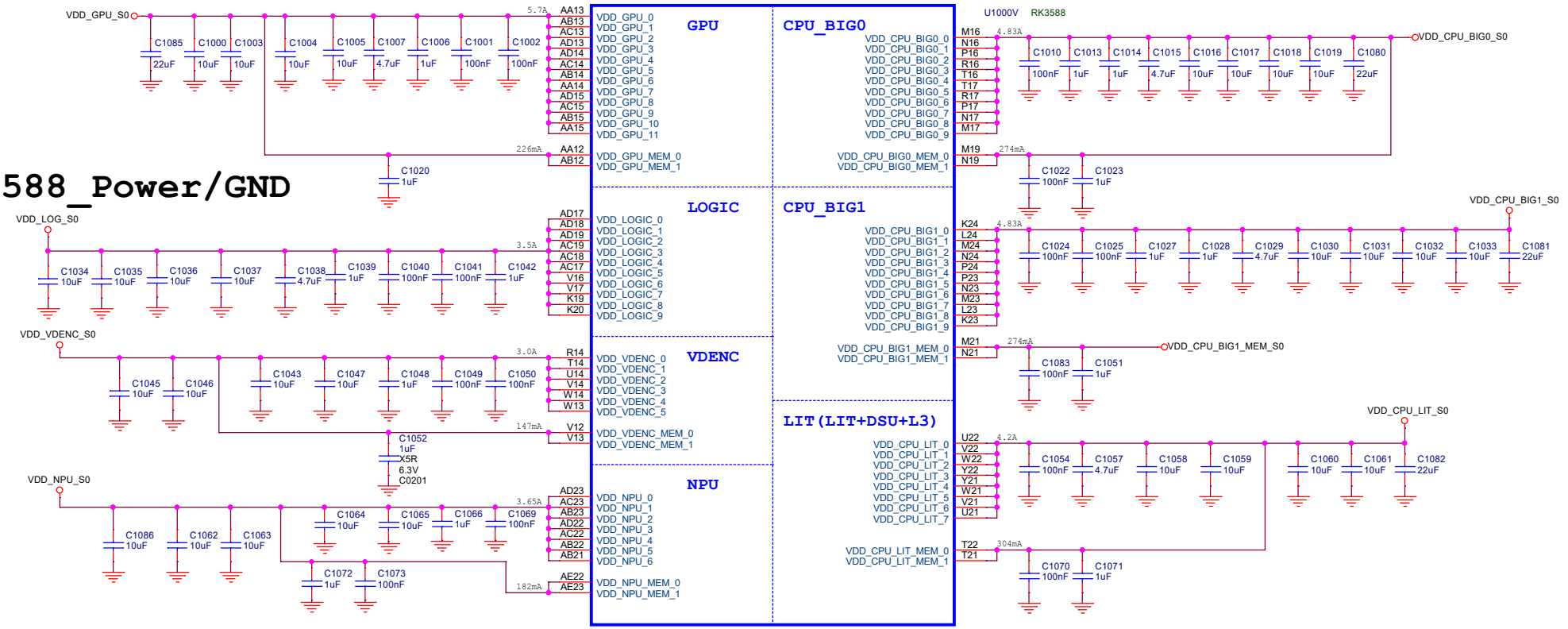
| Size | Title: | REV |
|---|---|---|
| A4 | ROCK 5T | 1.2 |
| | Page Name: Table of Content | |
| Date: Thursday, January 09, 2025 | Sheet 2 of 36 | |

# Block Diagram

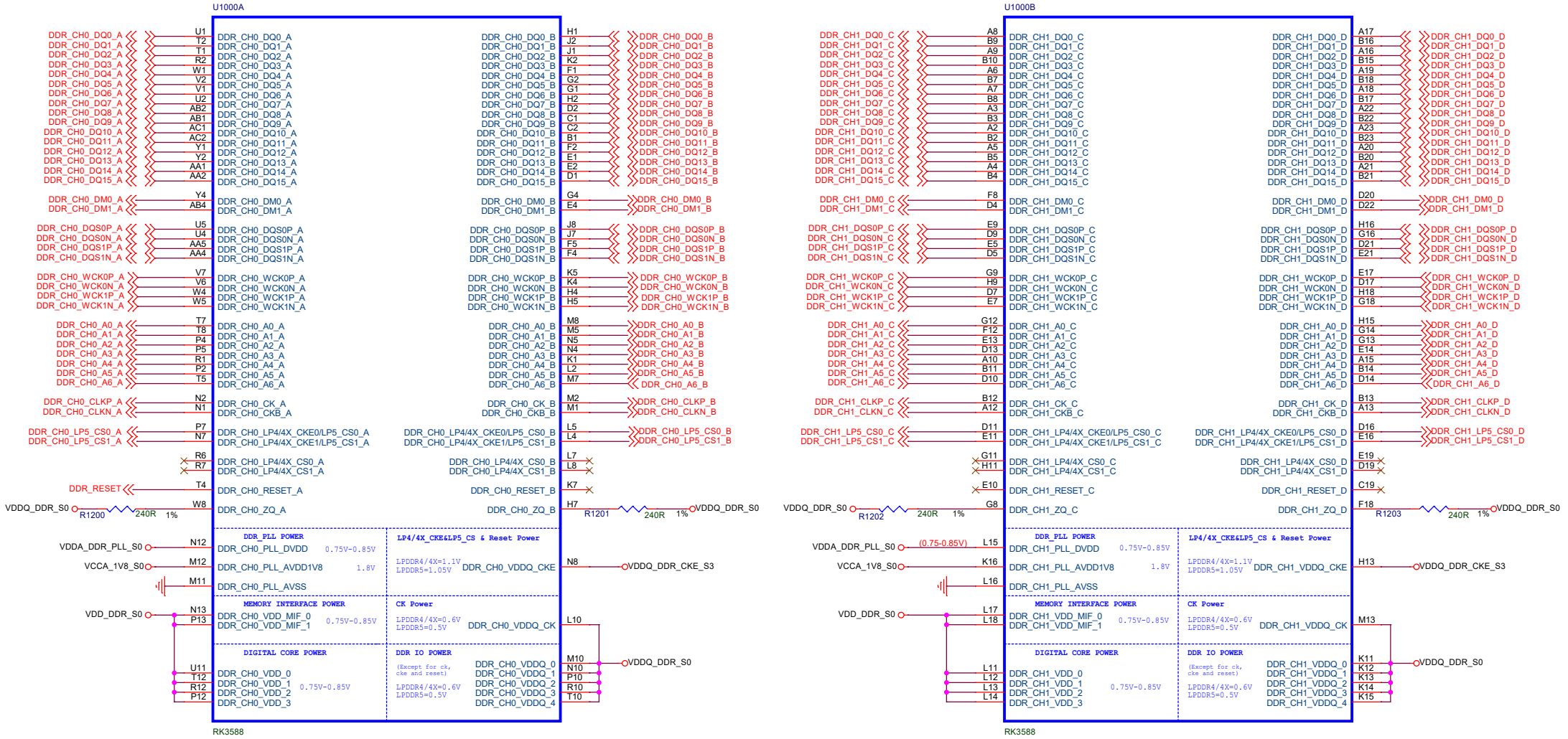



| | | | |
|---|---|---|---|
| Size | Title: | ROCK 5T | REV |
| A2 | Page Name: | Block Diagram | 1.2 |
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# RK3588_Power/GND

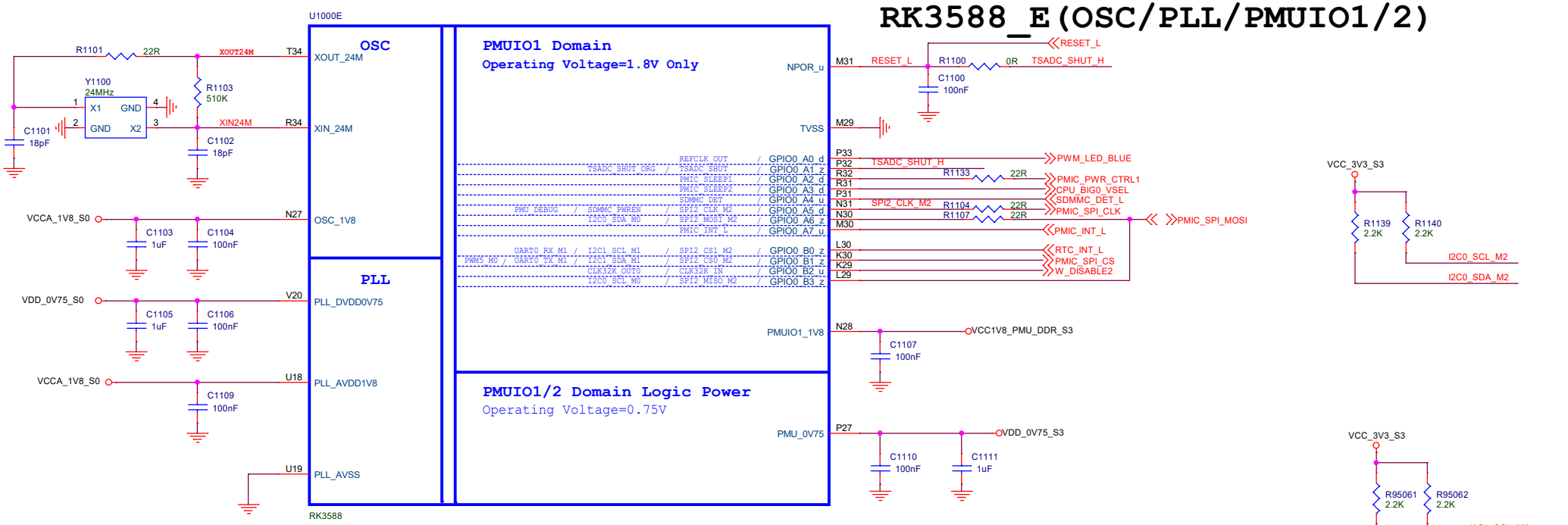


# RK3588 DDR Controler

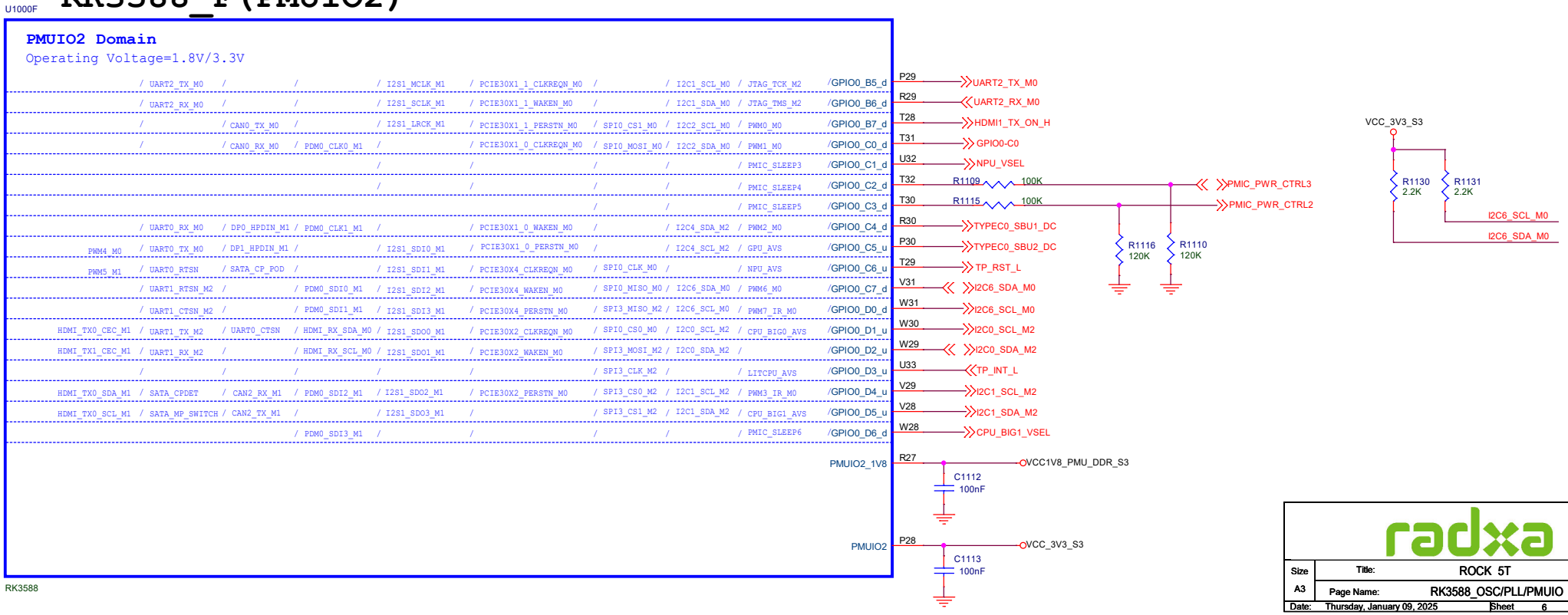



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|---|---|---|
| Size: A3 | Title: ROCK 5T | REV: 1.2 |
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# RK3588_E(OSC/PLL/PMUIO1/2)

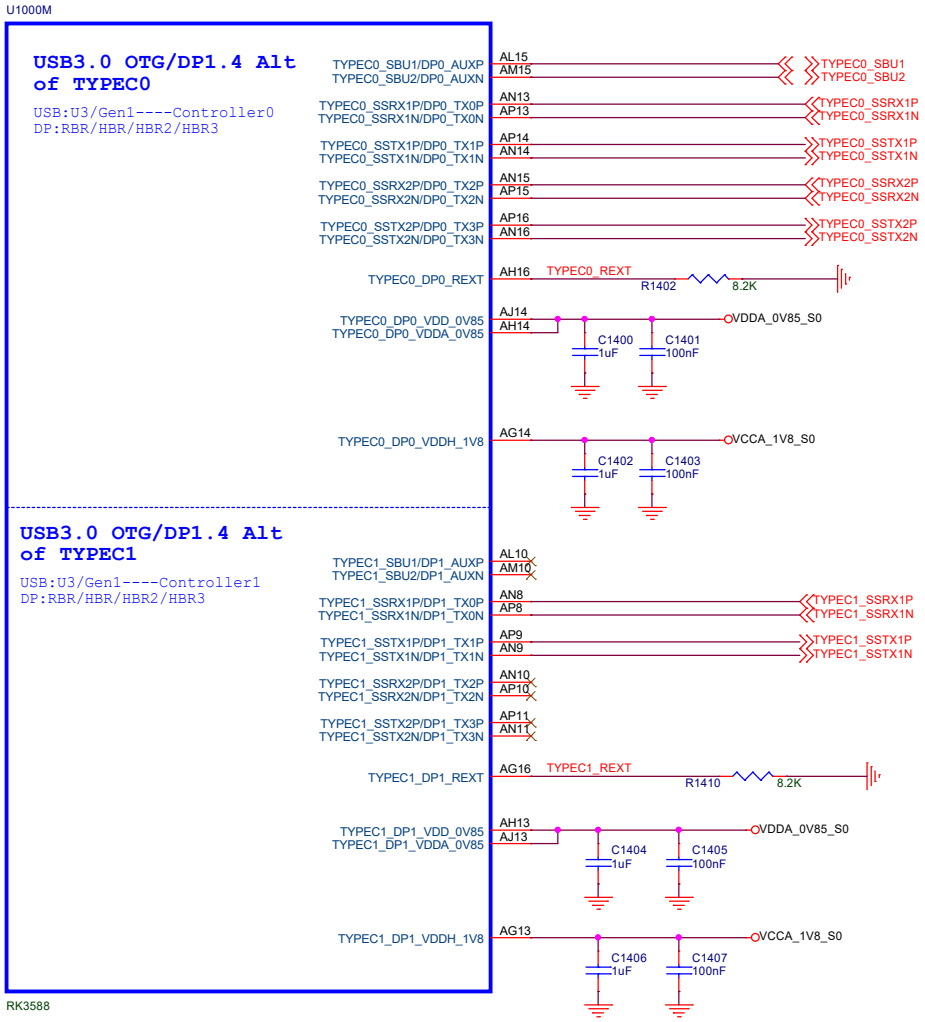



# RK3588_F(PMUIO2)

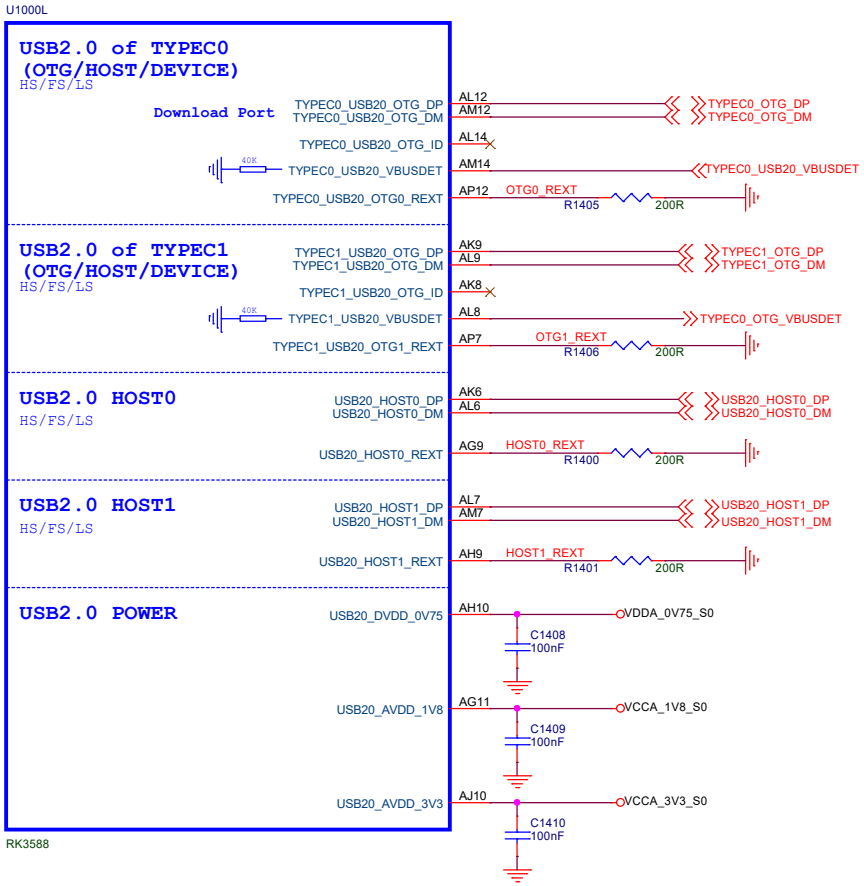



| | | |
|---|---|---|
| Size A3 | Title: ROCK 5T | REV 1.2 |
| | Page Name: RK3588_OSC/PLL/PMUIO | |
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# RK3588_M(TYPEC/DP)

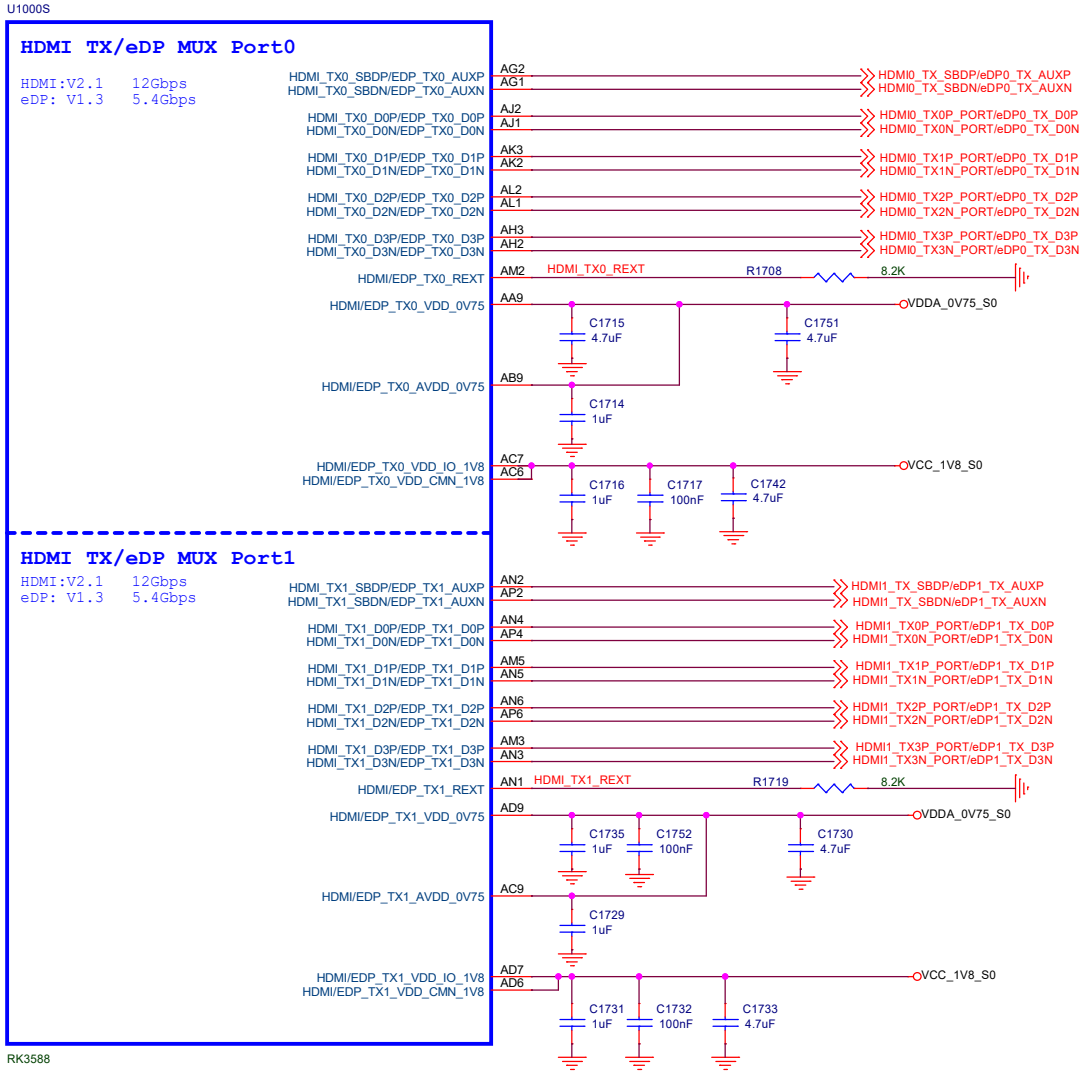



# RK3588_L(USB2.0 HOST/OTG)

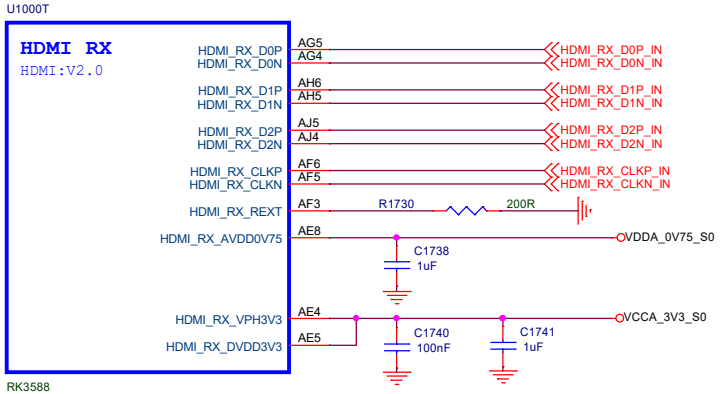



| | | |
|---|---|---|
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# RK3588_S(HDMI2.1 TX)




# RK3588_T(HDMI20 RX)




| | | |
|---|---|---|
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| | Page Name: RK3588_HDMI/eDP Interface | |
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# RK3588_C(EMMC)

U1000C

**EMMCIO Domain**

Operating Voltage=1.8V

| | | | |
|---|---|---|---|
| | FSPI_D0_M0 / EMMC_D0 | / GPIO2_D0_u | Y33 |
| | FSPI_D1_M0 / EMMC_D1 | / GPIO2_D1_u | W33 |
| | FSPI_D2_M0 / EMMC_D2 | / GPIO2_D2_u | V32 |
| | FSPI_D3_M0 / EMMC_D3 | / GPIO2_D3_u | AA33 |
| UART5_RX_M2 | / I2C1_SCL_M3 / EMMC_D4 | / GPIO2_D4_u | Y32 |
| UART5_TX_M2 | / I2C1_SDA_M3 / EMMC_D5 | / GPIO2_D5_u | AA32 |
| | FSPI_CS0N_M0/ EMMC_D6 | / GPIO2_D6_u | W32 |
| | FSPI_CS1N_M0/ EMMC_D7 | / GPIO2_D7_u | V33 |
| | FSPI_CLK_M0 / EMMC_CMD | / GPIO2_A0_u | W34 |
| | / EMMC_CLKOUT | / GPIO2_A1_d | V34 |
| UART5_CTSN_M1 | / I2C2_SDA_M2 / EMMC_DATA_STROBE | / GPIO2_A2_d | Y34 |
| UART5_RTSN_M1 | / I2C2_SCL_M2 / EMMC_RSTN | / GPIO2_A3_d | AA34 |
| | | EMMCIO_1V8 | V26 |

Y33 → eMMC_D0; W33 → eMMC_D1; V32 → eMMC_D2; AA33 → eMMC_D3; Y32 → eMMC_D4; AA32 → eMMC_D5; W32 → eMMC_D6; V33 → eMMC_D7

W34 → eMMC_CMD

V34 → R1317 0R → eMMC_CLKOUT

Y34 → eMMC_DATA_STROBE

AA34 → eMMC_RSTn

C145 NC

V26 → VCC_1V8_S3

C1300 1uF

RK3588

# RK3588_D(VCCIO2 )

U1000D

**VCCIO2 Domain**

Operating Voltage=1.8V/3.3V

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| PWM8_M1 | / I2C3_SCL_M4 | / PDM1_SDI3_M0 | / JTAG_TCK_M1 | / UART2_TX_M1 | / SDMMC_D0 | / GPIO4_D0_u | AD2 |
| PWM9_M1 | / I2C3_SDA_M4 | / PDM1_SDI2_M0 | / JTAG_TMS_M1 | / UART2_RX_M1 | / SDMMC_D1 | / GPIO4_D1_u | AD1 |
| | / I2C8_SCL_M0 | / PDM1_SDI1_M0 | / JTAG_TCK_M0 | / UART5_CTSN_M0 | / SDMMC_D2 | / GPIO4_D2_u | AF2 |
| PWM10_M1 | / I2C8_SDA_M0 | / PDM1_SDI0_M0 | / JTAG_TMS_M0 | / UART5_RTSN_M0 | / SDMMC_D3 | / GPIO4_D3_u | AF1 |
| PWM7_IR_M1 | / CAN0_TX_M1 | / PDM1_CLK1_M0 | / MCU_JTAG_TCK_M0 | / UART5_RX_M0 | / SDMMC_CMD | / GPIO4_D4_u | AE2 |
| TEST_CLKOUT_M0 | / CAN0_RX_M1 | / PDM1_CLK0_M0 | / MCU_JTAG_TMS_M0 | / UART5_TX_M0 | / SDMMC_CLK | / GPIO4_D5_d | AE1 |
| | | | | | | VCCIO2_1V8 | AA7 |
| | | | | | | VCCIO2 | Y7 |

AD2 → SDMMC0_D0; AD1 → SDMMC0_D1; AF2 → SDMMC_D2/JTAG_TCK_M0; AF1 → SDMMC_D3/JTAG_TMS_M0; AE2 → SDMMC_CMD/MCU_JTAG_TCK_M0; AE1 → SDMMC_CLK/MCU_JTAG_TMS_M0

AA7 → VCC_1V8_S0

C1301 100nF

Y7 → VCCIO_SD_S0

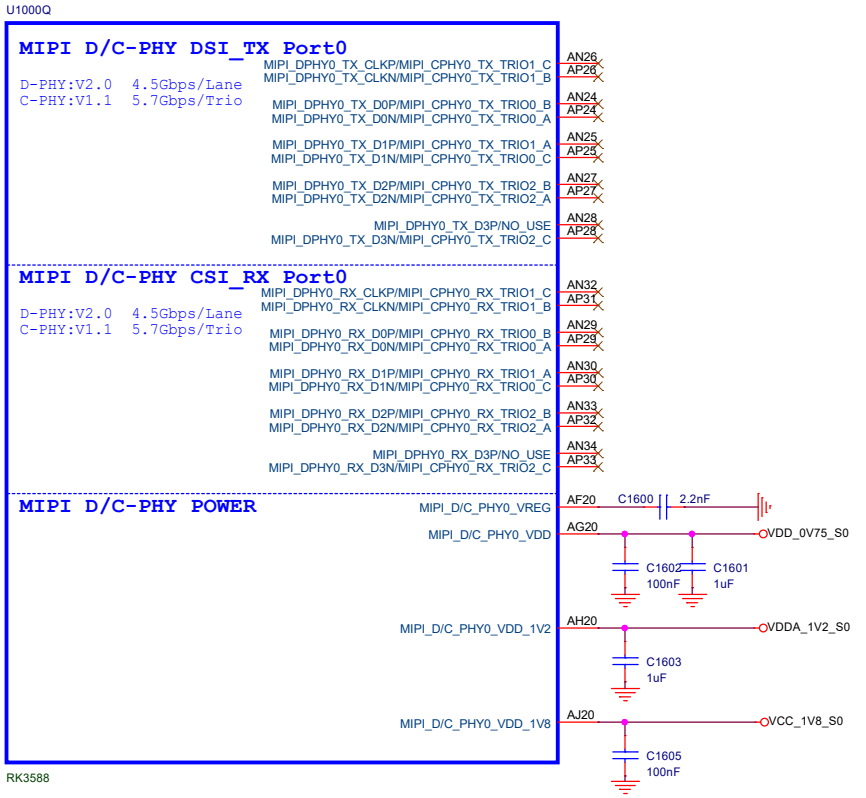
C1302 100nF

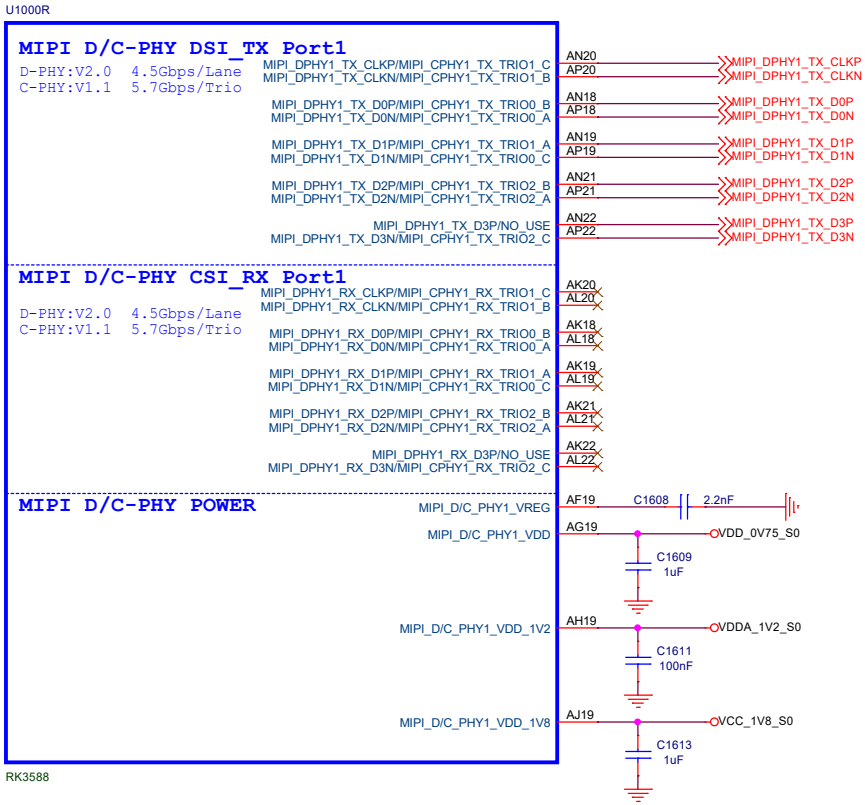
RK3588

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| Size | Title: | ROCK 5T | REV |
|---|---|---|---|
| A4 | Page Name: | RK3588_Flash/SD Controller | 1.2 |
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# RK3588_Q/R(MIPI_D/C_PHY0/1)




| | | |
|---|---|---|
| Size: A3 | Title: ROCK 5T | REV: 1.2 |
| | Page Name: RK3588_MIPI/DSI/CSI | |
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# RK3588_P(MIPI_CSI_RX_PHY)

U1000P

## MIPI DPHY CSI_RX Port0

MIPI V1.2/2.5Gbps

| | | |
|---|---|---|
| MIPI_CSI0_CLK0P | AJ33 | MIPI_CSI0_RX_CLK0P |
| MIPI_CSI0_CLK0N | AJ34 | MIPI_CSI0_RX_CLK0N |
| MIPI_CSI0_D0P | AG33 | MIPI_CSI0_RX_D0P |
| MIPI_CSI0_D0N | AG34 | MIPI_CSI0_RX_D0N |
| MIPI_CSI0_D1P | AH33 | MIPI_CSI0_RX_D1P |
| MIPI_CSI0_D1N | AH34 | MIPI_CSI0_RX_D1N |
| MIPI_CSI0_CLK1P | AM33 | MIPI_CSI0_RX_CLK1P |
| MIPI_CSI0_CLK1N | AM34 | MIPI_CSI0_RX_CLK1N |
| MIPI_CSI0_D2P | AK33 | MIPI_CSI0_RX_D2P |
| MIPI_CSI0_D2N | AK34 | MIPI_CSI0_RX_D2N |
| MIPI_CSI0_D3P | AL33 | MIPI_CSI0_RX_D3P |
| MIPI_CSI0_D3N | AL34 | MIPI_CSI0_RX_D3N |

MIPI_CSI0_AVCC0V75 — AB25 — VDD_0V75_S0

C1616 100nF, C1615 1uF

MIPI_CSI0_AVCC1V8 — AB26 — VCC_1V8_S0

C1617 1uF

## MIPI DPHY CSI_RX Port1

MIPI V1.2/2.5Gbps

| | | |
|---|---|---|
| MIPI_CSI1_CLK0P | AJ31 | MIPI_CSI1_RX_CLK0P |
| MIPI_CSI1_CLK0N | AJ32 | MIPI_CSI1_RX_CLK0N |
| MIPI_CSI1_D0P | AG31 | MIPI_CSI1_RX_D0P |
| MIPI_CSI1_D0N | AG32 | MIPI_CSI1_RX_D0N |
| MIPI_CSI1_D1P | AH31 | MIPI_CSI1_RX_D1P |
| MIPI_CSI1_D1N | AH32 | MIPI_CSI1_RX_D1N |
| MIPI_CSI1_CLK1P | AM31 | MIPI_CSI1_RX_CLK1P |
| MIPI_CSI1_CLK1N | AM32 | MIPI_CSI1_RX_CLK1N |
| MIPI_CSI1_D2P | AK31 | MIPI_CSI1_RX_D2P |
| MIPI_CSI1_D2N | AK32 | MIPI_CSI1_RX_D2N |
| MIPI_CSI1_D3P | AL31 | MIPI_CSI1_RX_D3P |
| MIPI_CSI1_D3N | AL32 | MIPI_CSI1_RX_D3N |

MIPI_CSI1_AVCC0V75 — AA25 — VDD_0V75_S0

C1619 1uF

MIPI_CSI1_AVCC1V8 — AA26 — VCC_1V8_S0

C1621 100nF

RK3588

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| | | | |
|---|---|---|---|
| Size A4 | Title: | ROCK 5T | REV 1.2 |
| | Page Name: | RK3588_MIPI CSI | |
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# RK3588_N(PCIE20)

U1000N

## PCIE20/SATA30 Mux0

PCIe20x1_2 1Lane(RC)--4(1L2)

SATA30 HOST Controller0

MUX

input or output

PCIE20_0_REFCLKP / PCIE20_0_REFCLKN — L32 / L33 — PCIE20_0_REFCLKP / PCIE20_0_REFCLKN

PCIE20_0_TXP/SATA30_0_TXP / PCIE20_0_TXN/SATA30_0_TXN — M34 / M33 — PCIE20_0_TXP / PCIE20_0_TXN

PCIE20_0_RXP/SATA30_0_RXP / PCIE20_0_RXN/SATA30_0_RXN — N33 / N34 — PCIE20_0_RXP / PCIE20_0_RXN

PCIE20_SATA30_0_AVDD_0V85 — M28 — VDDA_0V85_S0 — C1810 10uF

PCIE20_SATA30_0_AVDD_1V8 — M27 — VCCA_1V8_S0 — C1808 1uF

PCIE20:Gen1/Gen2
SATA30:Gen1/Gen2/Gen3

## PCIE20/SATA30 Mux1

PCIe30x1_0 1Lane(RC)--2(1L0)

SATA30 HOST Controller1

MUX

input or output

PCIE20_1_REFCLKP / PCIE20_1_REFCLKN — H32 / H33 — PCIE20_1_REFCLKP / PCIE20_1_REFCLKN

PCIE20_1_TXP/SATA30_1_TXP / PCIE20_1_TXN/SATA30_1_TXN — K33 / K34 — PCIE20_1_TXP / PCIE20_1_TXN

PCIE20_1_RXP/SATA30_1_RXP / PCIE20_1_RXN/SATA30_1_RXN — J33 / J34 — PCIE20_1_RXP / PCIE20_1_RXN

PCIE20_SATA30_1_AVDD_0V85 — L28 — VDDA_0V85_S0 — C1818 100nF

PCIE20_SATA30_1_AVDD_1V8 — L27 — VCCA_1V8_S0 — C1816 100nF

PCIE20:Gen1/Gen2
SATA30:Gen1/Gen2/Gen3

## PCIE20/SATA30/USB30 HOST Mux2

PCIe30x1_1 1Lane(RC)--3(1L1)

SATA30 HOST Controller2

USB30 HOST Controller2

MUX

input or output

PCIE20_2_REFCLKP / PCIE20_2_REFCLKN — G31 / G30 — PCIE20_2_REFCLKP / PCIE20_2_REFCLKN

PCIE20_2_TXP/SATA30_2_TXP/USB30_2_SSTXP / PCIE20_2_TXN/SATA30_2_TXN/USB30_2_SSTXN — H30 / H29 — PCIE20_2_TXP / PCIE20_2_TXN

PCIE20_2_RXP/SATA30_2_RXP/USB30_2_SSRXP / PCIE20_2_RXN/SATA30_2_RXN/USB30_2_SSRXN — J31 / J30 — PCIE20_2_RXP / PCIE20_2_RXN

PCIE20_SATA30_USB30_2_AVDD_0V85 — K28 — VDDA_0V85_S0 — C1806 1uF

PCIE20_SATA30_USB30_2_AVDD_1V8 — K27 — VCCA_1V8_S0 — C1804 1uF

PCIE20:Gen1/Gen2
USB 30:Gen1
SATA30:Gen1/Gen2/Gen3

RK3588

# RK3588_O(PCIE30)

U1000O

## PCIE30X4

PCIe30X4 4Lane(DM)--0(4L)

PCIe30x1_0 1Lane(RC)--2(1L0)

PCIE20/SATA30 MUX1

PCIE30:
PCIe1.1(2.5Gbps)
PCIe2.1(5Gbps)
PCIe3.0(8Gbps)

PCIe30x2 2Lane(RC)--1(2L)

PCIe30x1_1 1Lane(RC)--3(1L1)

PCIE20/SATA30/USB30 MUX2

## PCIE30 PORT0

input PCIE30_PORT0_REF_CLKP / PCIE30_PORT0_REF_CLKN — E33 / E34 — PCIE30_PORT0_REFCLKP_IN / PCIE30_PORT0_REFCLKN_IN

Lane0

PCIE30_PORT0_TX0P / PCIE30_PORT0_TX0N — D32 / D33 — PCIE30_PORT0_TX0P / PCIE30_PORT0_TX0N

PCIE30_PORT0_RX0P / PCIE30_PORT0_RX0N — G33 / G34 — PCIE30_PORT0_RX0P / PCIE30_PORT0_RX0N

Lane1

PCIE30_PORT0_TX1P / PCIE30_PORT0_TX1N — C33 / C34 — PCIE30_PORT0_TX1P / PCIE30_PORT0_TX1N

PCIE30_PORT0_RX1P / PCIE30_PORT0_RX1N — F32 / F33 — PCIE30_PORT0_RX1P / PCIE30_PORT0_RX1N

PCIE30_PORT0_RESREF — B34 — PCIE30_0_REF — R1801 200R

PCIE30_PORT0_AVDD0V75 — G24 — VDD_0V75_S0 — C1802 1uF — C1803 4.7uF

PCIE30_PORT0_AVDD1V8 — G23 — VCCA_1V8_S0 — C1800 1uF — C1801 4.7uF

## PCIE30 PORT1

input PCIE30_PORT1_REF_CLKP / PCIE30_PORT1_REF_CLKN — A28 / B28 — PCIE30_PORT1_REFCLKP_IN / PCIE30_PORT1_REFCLKN_IN

Lane0

PCIE30_PORT1_TX0P / PCIE30_PORT1_TX0N — B30 / A30 — PCIE30_PORT1_TX2P / PCIE30_PORT1_TX2N

PCIE30_PORT1_RX0P / PCIE30_PORT1_RX0N — B32 / A32 — PCIE30_PORT1_RX2P / PCIE30_PORT1_RX2N

Lane1

PCIE30_PORT1_TX1P / PCIE30_PORT1_TX1N — C29 / B29 — PCIE30_PORT1_TX3P / PCIE30_PORT1_TX3N

PCIE30_PORT1_RX1P / PCIE30_PORT1_RX1N — C31 / B31 — PCIE30_PORT1_RX3P / PCIE30_PORT1_RX3N

PCIE30_PORT1_RESREF — A33 — PCIE30_1_REF — R1807 200R

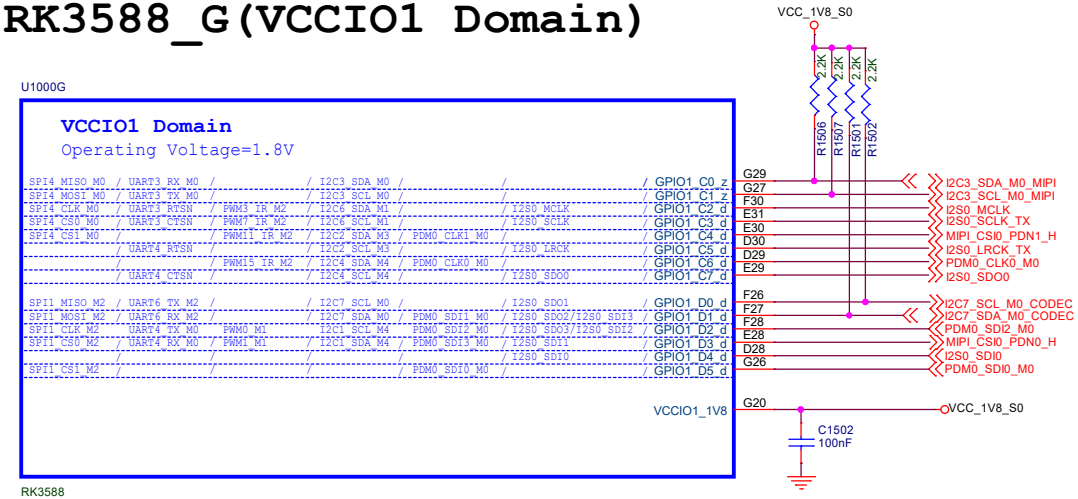
PCIE30_PORT1_AVDD0V75 — H24 — VDD_0V75_S0 — C1814 1uF — C1815 4.7uF

PCIE30_PORT1_AVDD1V8 — H23 — VCCA_1V8_S0 — C1812 100nF — C1813 4.7uF

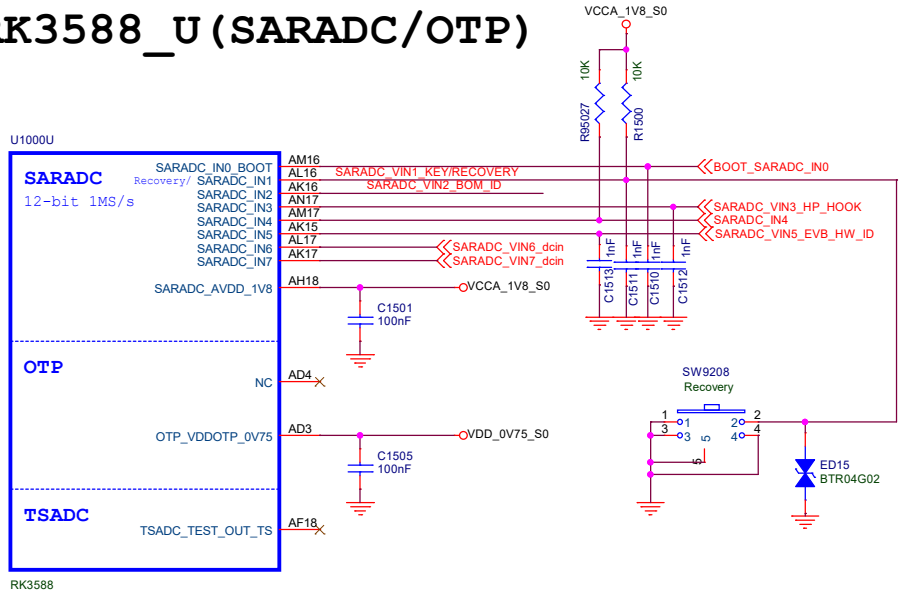
RK3588

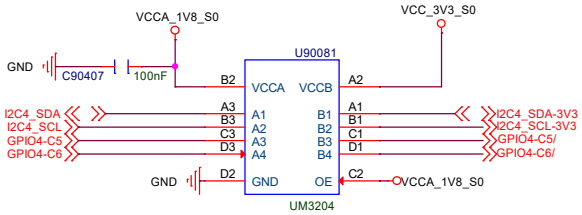
| | | | |
|---|---|---|---|
| Size A3 | Title: ROCK 5T | | REV 1.2 |
| | Page Name: RK3588_PCIE30/PCIE20/SATA30 | | |
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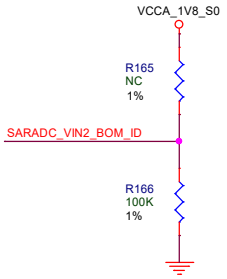
# RK3588_G(VCCIO1 Domain)




# RK3588_U(SARADC/OTP)




## BOOT MODE CONFIG

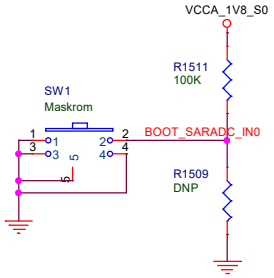
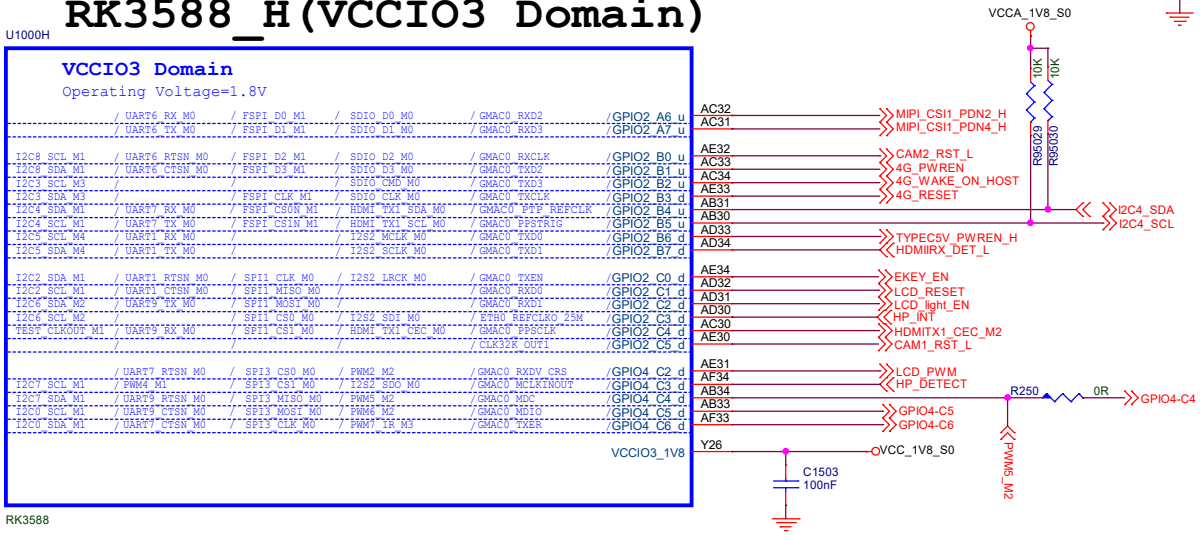



*TABLE 1*

| Item | Rup | Rdown | ADC | VOL | BOOT MODE |
|---|---|---|---|---|---|
| LEVEL1 | DNP | 100K | 0 | 0V | USB (Maskrom mode) |
| LEVEL2 | 100K | 20K | 682 | 0.3V | SD Card-USB |
| LEVEL3 | 100K | 51K | 1365 | 0.6V | EMMC-USB |
| LEVEL4 | 100K | 100K | 2047 | 0.9V | FSPI M0-USB |
| LEVEL5 | 100K | 200K | 2730 | 1.2V | FSPI M1-USB |
| LEVEL6 | 100K | 499K | 3412 | 1.5V | FSPI M2-USB |
| LEVEL7 | 100K | DNP | 4095 | 1.8V | FSPI_M2-FSPI_M1-FSPI_M0 -EMMC-SD Card-USB |

# RK3588_H(VCCIO3 Domain)




## BOARD ID CONFIG

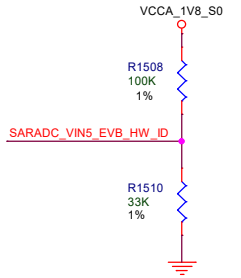


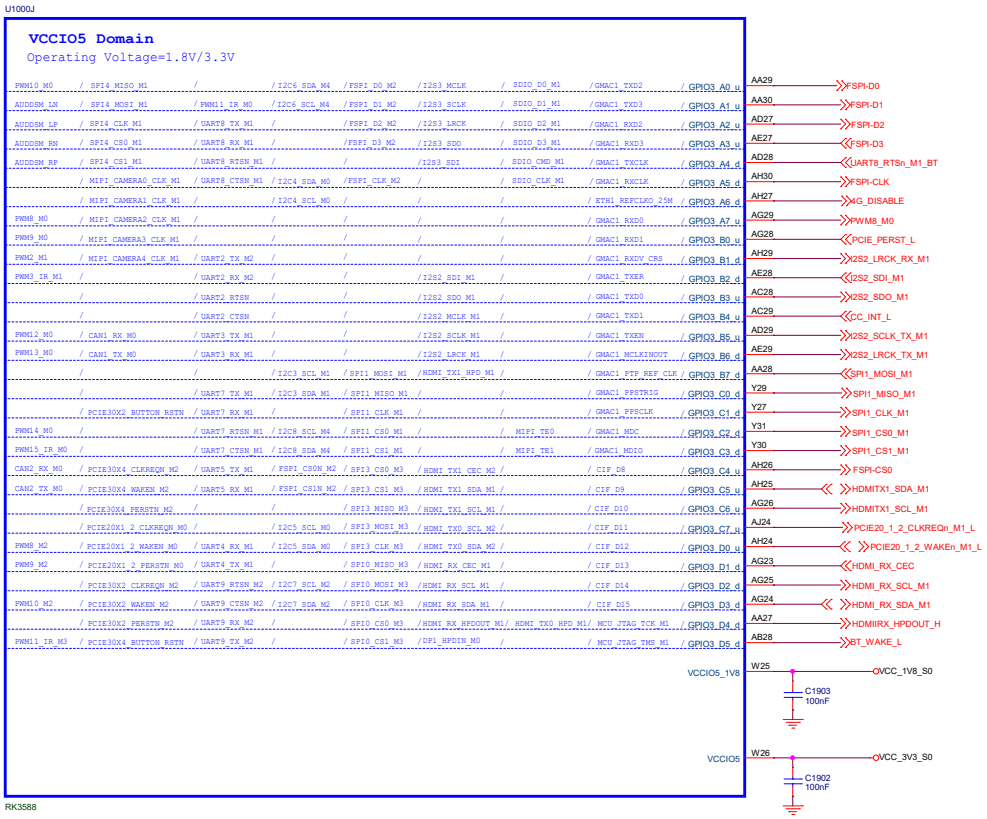

*TABLE 2*

| Item | Rup | Rdown | ADC | VOL | VERSION |
|---|---|---|---|---|---|
| LEVEL1 | DNP | 100K | 0 | 0V | A |
| LEVEL2 | 100K | 20K | 682 | 0.3V | B |
| LEVEL3 | 100K | 51K | 1365 | 0.6V | C |
| LEVEL4 | 100K | 100K | 2047 | 0.9V | D |
| LEVEL5 | 100K | 200K | 2730 | 1.2V | E |
| LEVEL6 | 100K | 499K | 3412 | 1.5V | F |
| LEVEL7 | 100K | DNP | 4095 | 1.8V | H |

radxa

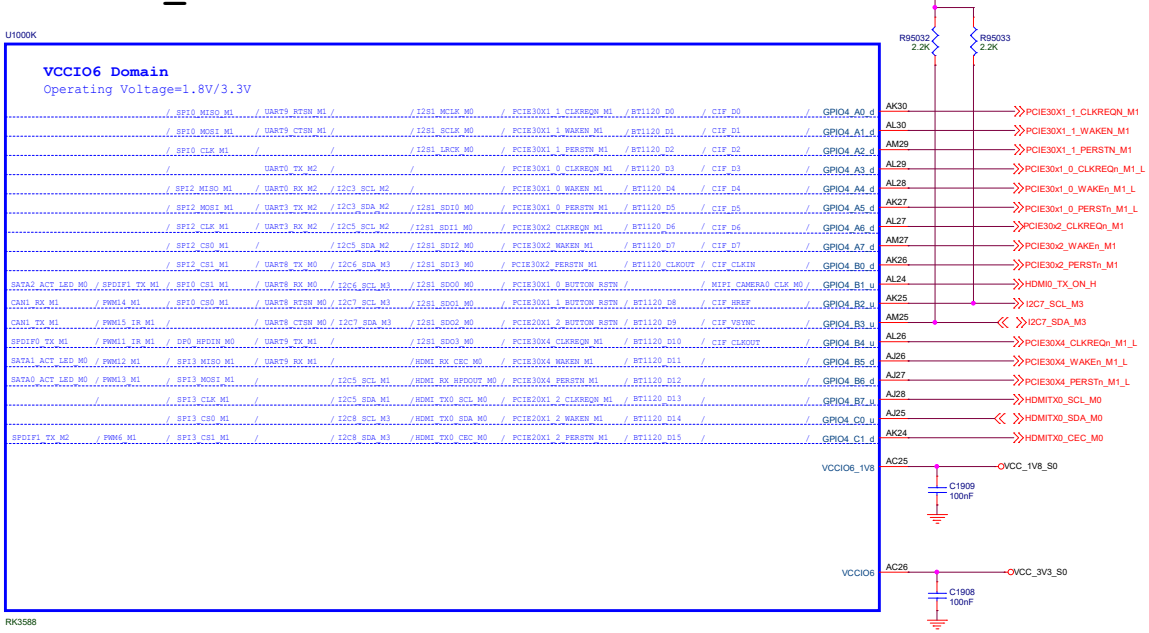
| Size | Title: | REV |
|---|---|---|
| A3 | ROCK 5T | |
| | Page Name: RK3588_SARADC_GPIO | 1.2 |
| Date: Thursday, January 09, 2025 | Sheet 13 of 36 | |

# RK3588_J(VCCIO5)



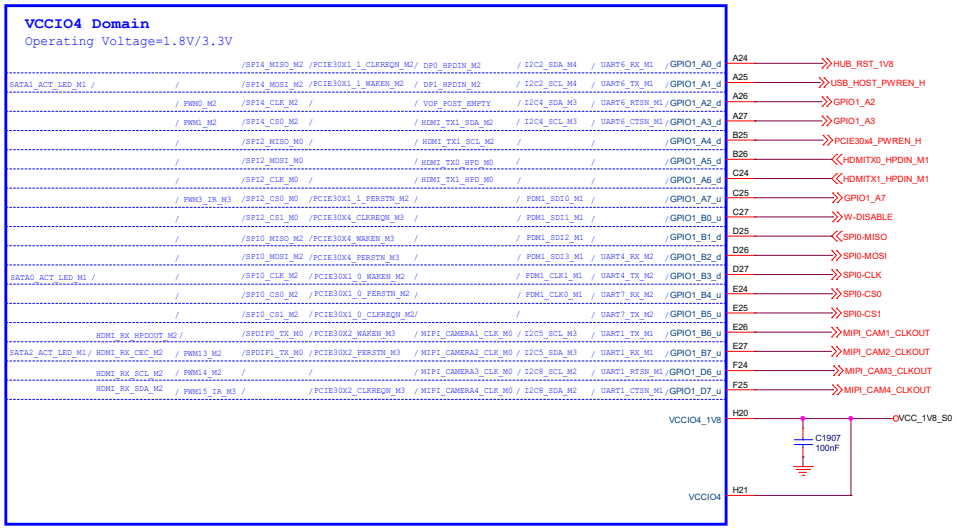

RK3588

# RK3588_K(VCCIO6)




RK3588

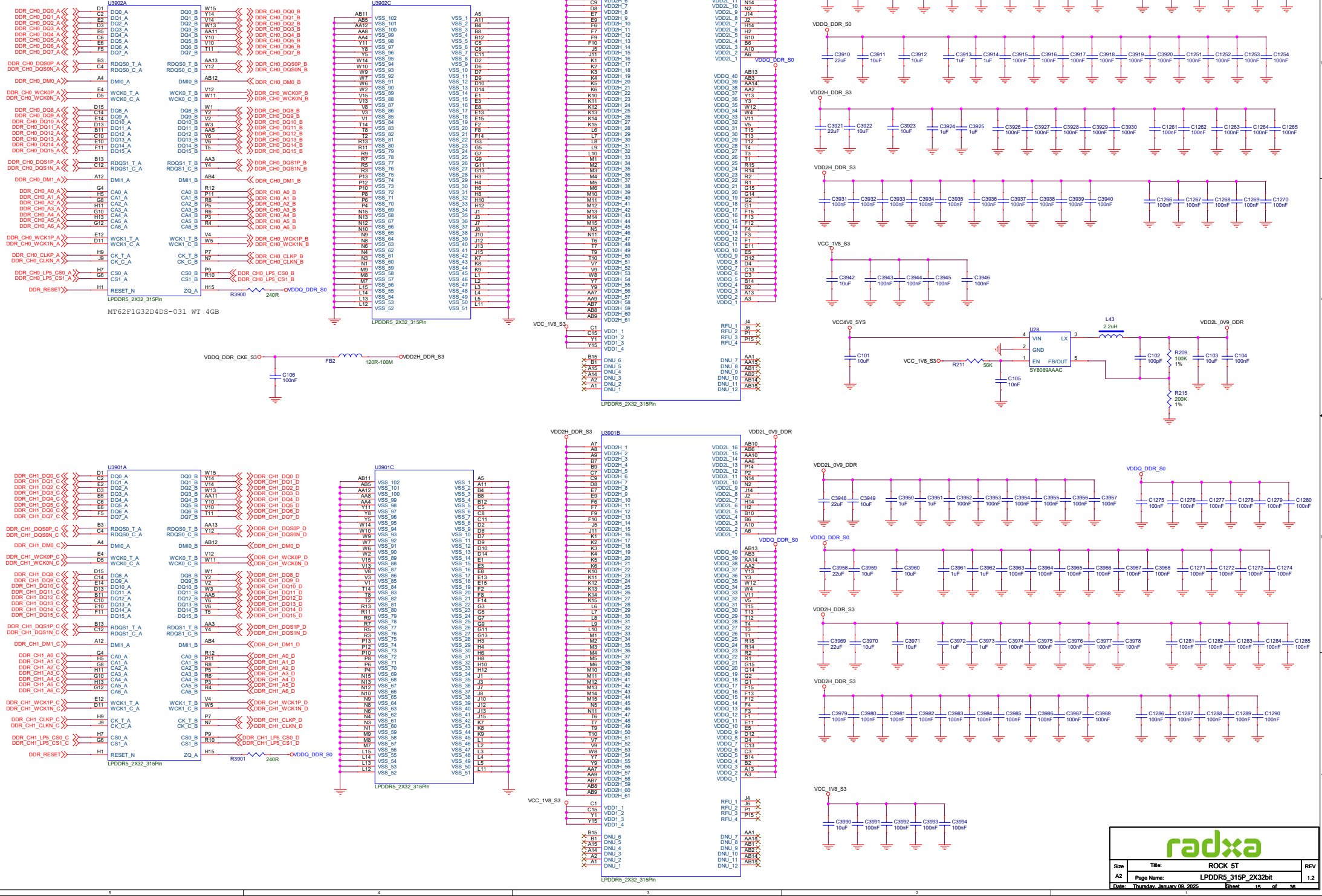
# RK3588_I(VCCIO4)




RK3588

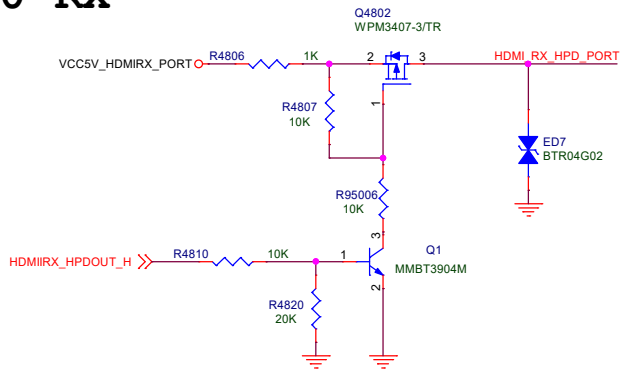
| | | |
|---|---|---|
| Size: A2 | Title: ROCK 5T | REV: 1.2 |
| | Page Name: RK3588_GPIO | |
| Date: Thursday, January 09, 2025 | Sheet 14 of 36 | |

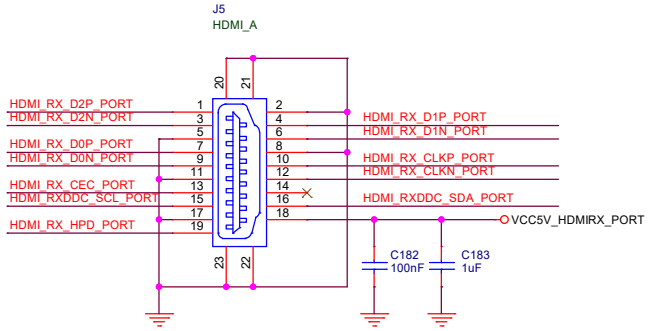
# LPDDR5

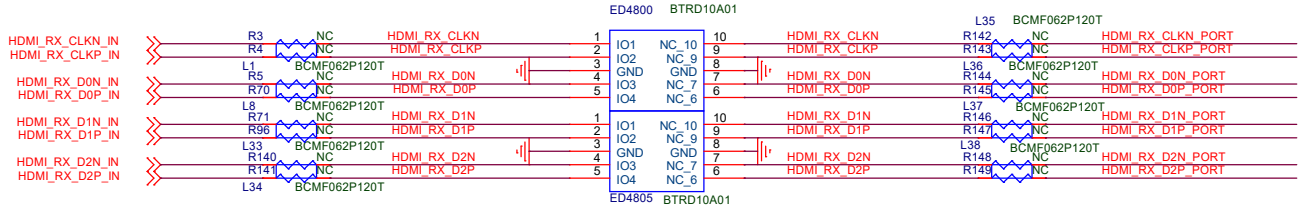


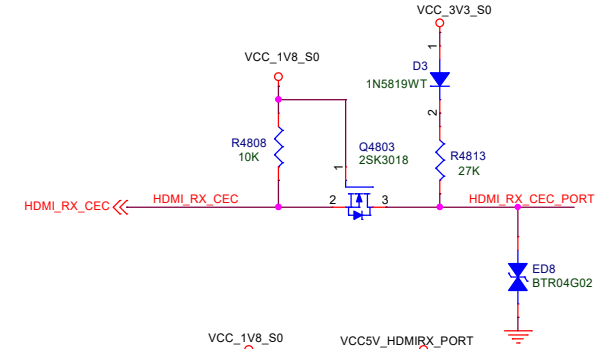

| | | | |
|---|---|---|---|
| Size | Title: ROCK_5T | | REV |
| A2 | Page Name: LPDDR5_315P_2X32bit | | 1.2 |
| Date: Thursday, January 09, 2025 | Sheet 15 of 36 | | |

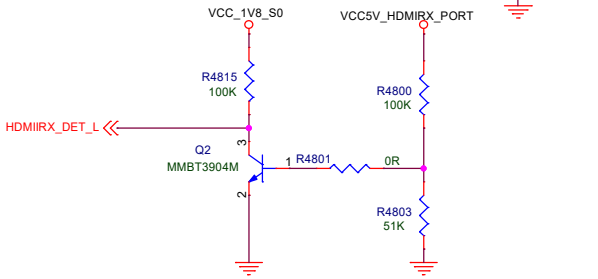
# HDMI2.0 RX

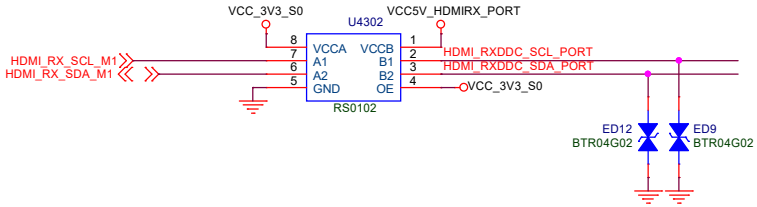


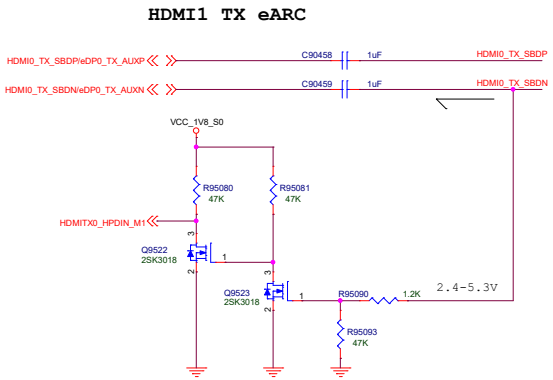



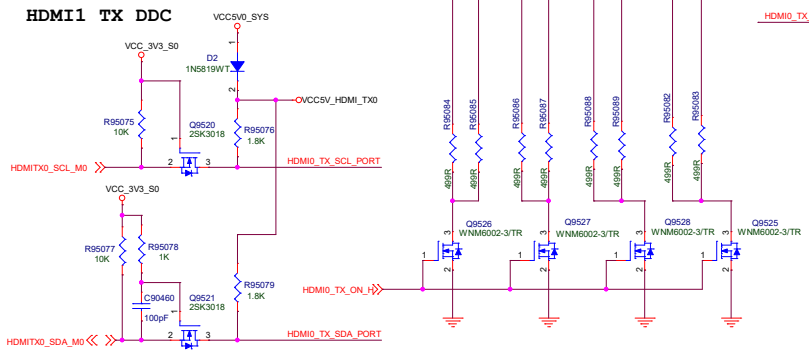



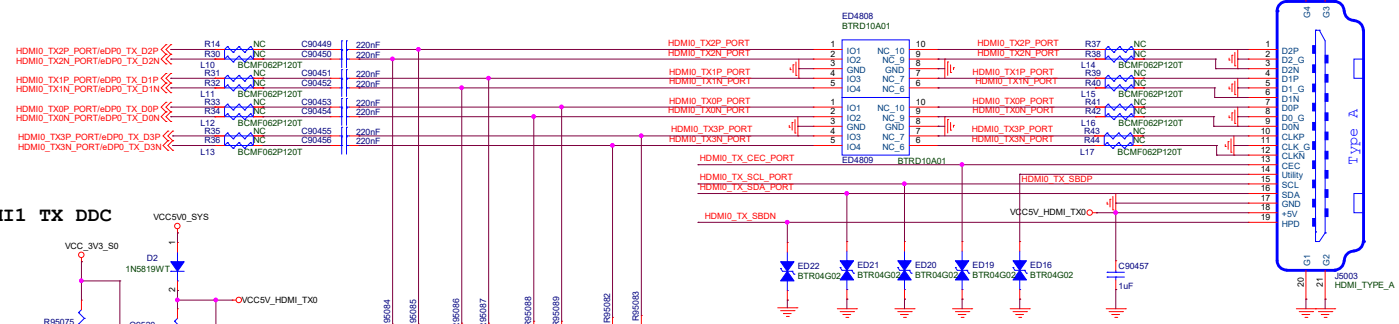



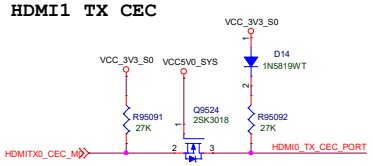

| | | | |
|---|---|---|---|
| Size | Title: ROCK 5T | | REV |
| A3 | Page Name: VI-HDMI2.0 RX | | 1.2 |
| Date: Thursday, January 09, 2025 | Sheet 16 of 36 | | |

# HDMI 1

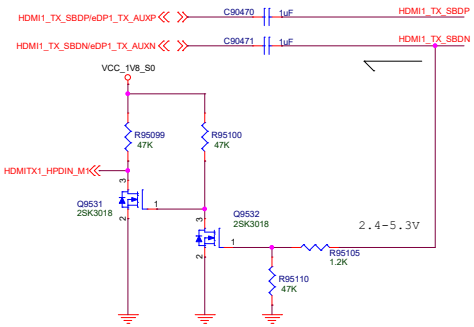


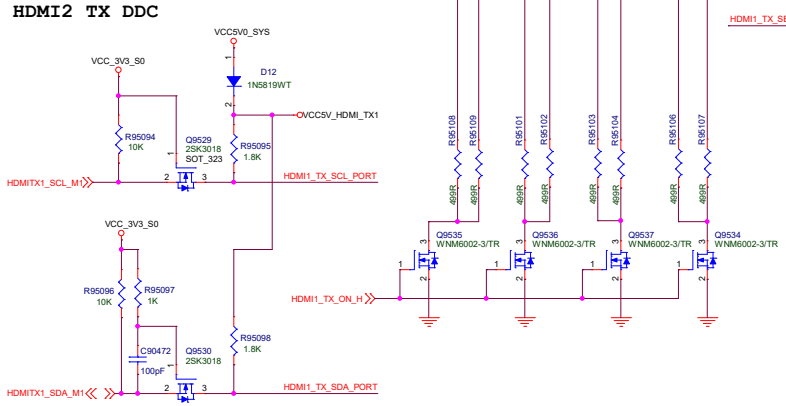



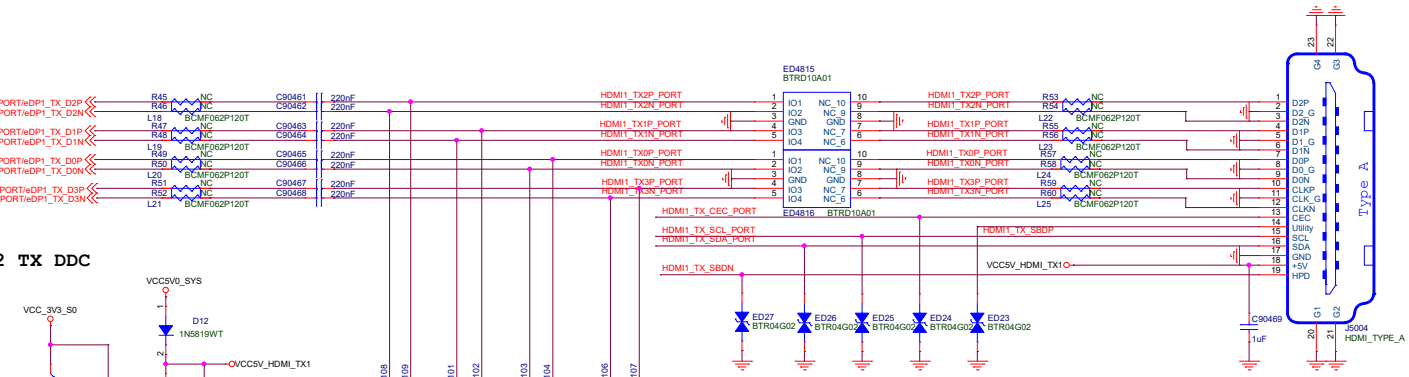



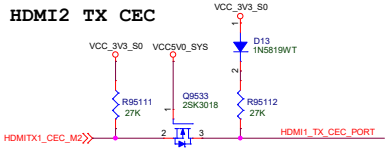




# HDMI 2



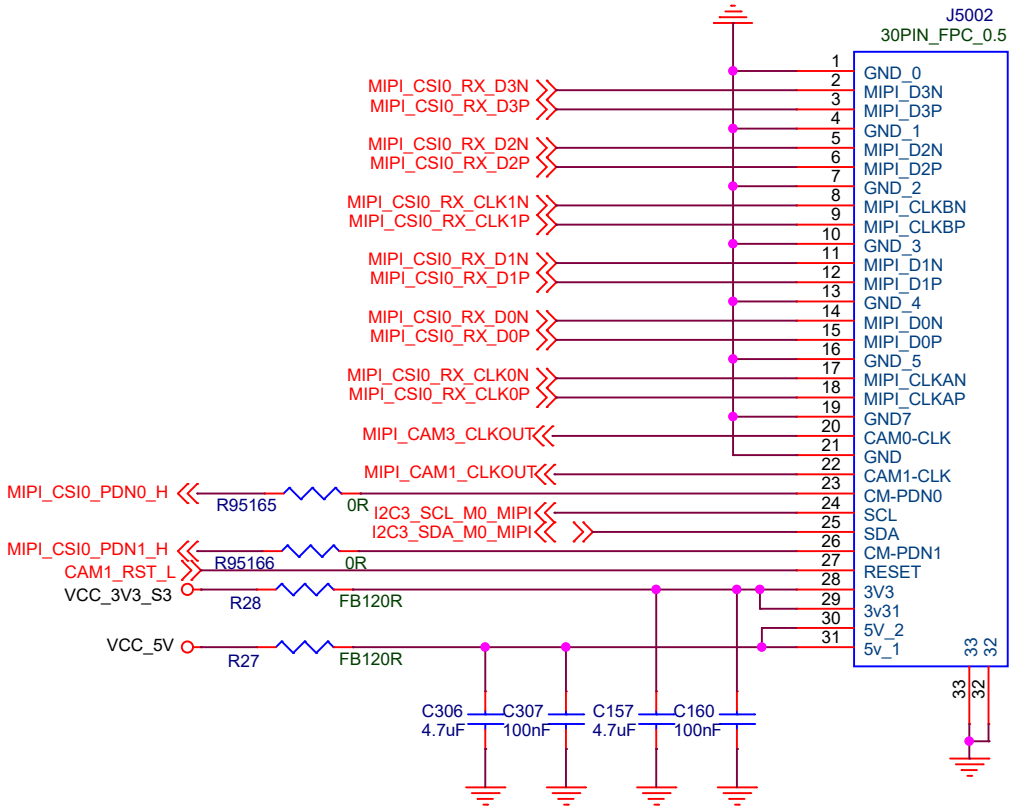



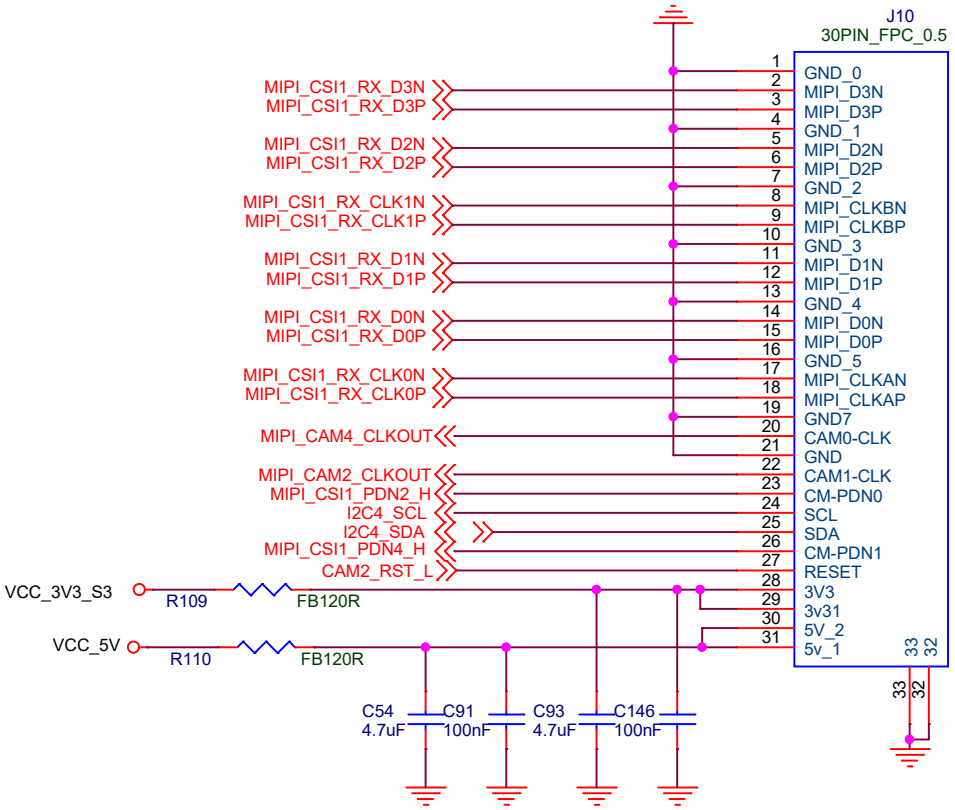

radxa

| Size | Title: | ROCK 5T | REV |
|---|---|---|---|
| A2 | Page Name: | VO-HDMI2.1 TX | 1.2 |
| Date: Thursday, January 09, 2025 | Sheet 17 of 36 | | |

# CAM1

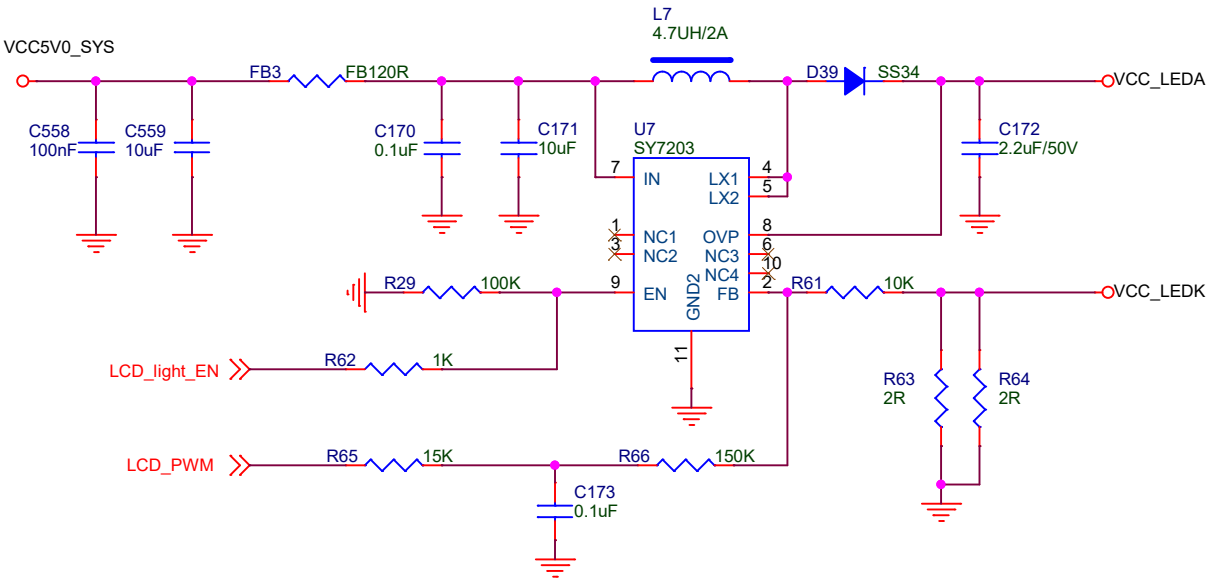


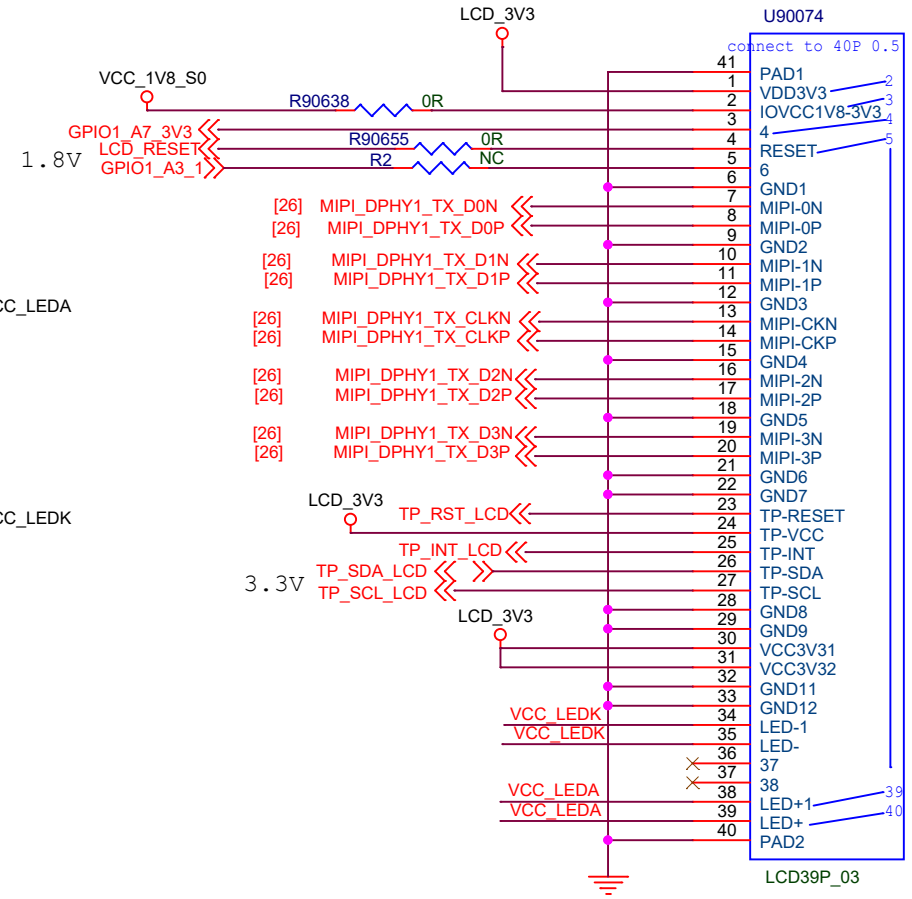

# CAM2



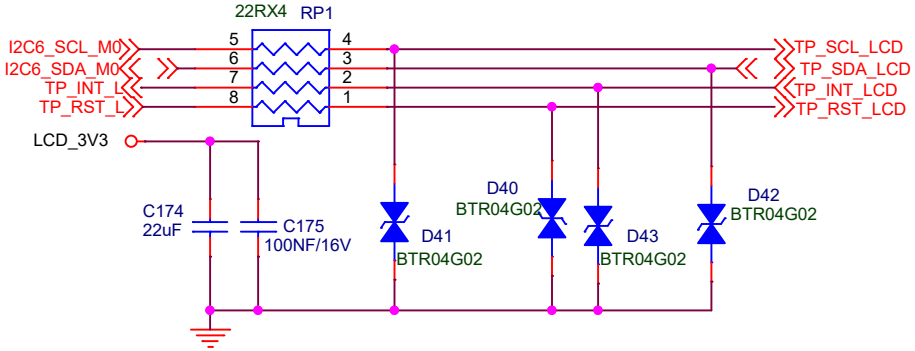

| | | |
|---|---|---|
| Size A4 | Title: ROCK 5T | REV 1.2 |
| | Page Name: Camera_MIPI-CSI | |
| Date: Thursday, January 09, 2025 | Sheet 18 of 36 | |

# MIPI DPHY1 TX



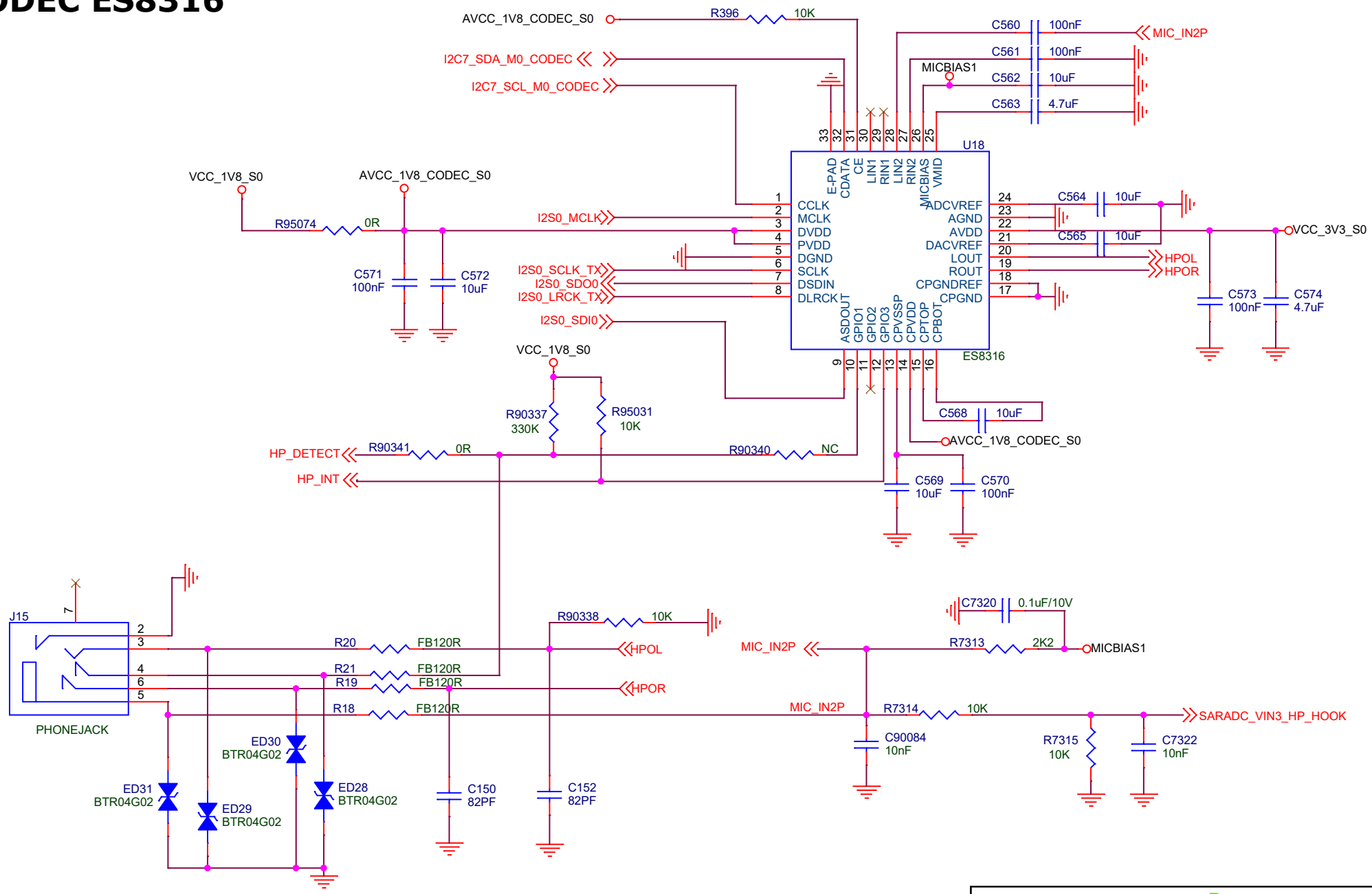




# LCD_TP



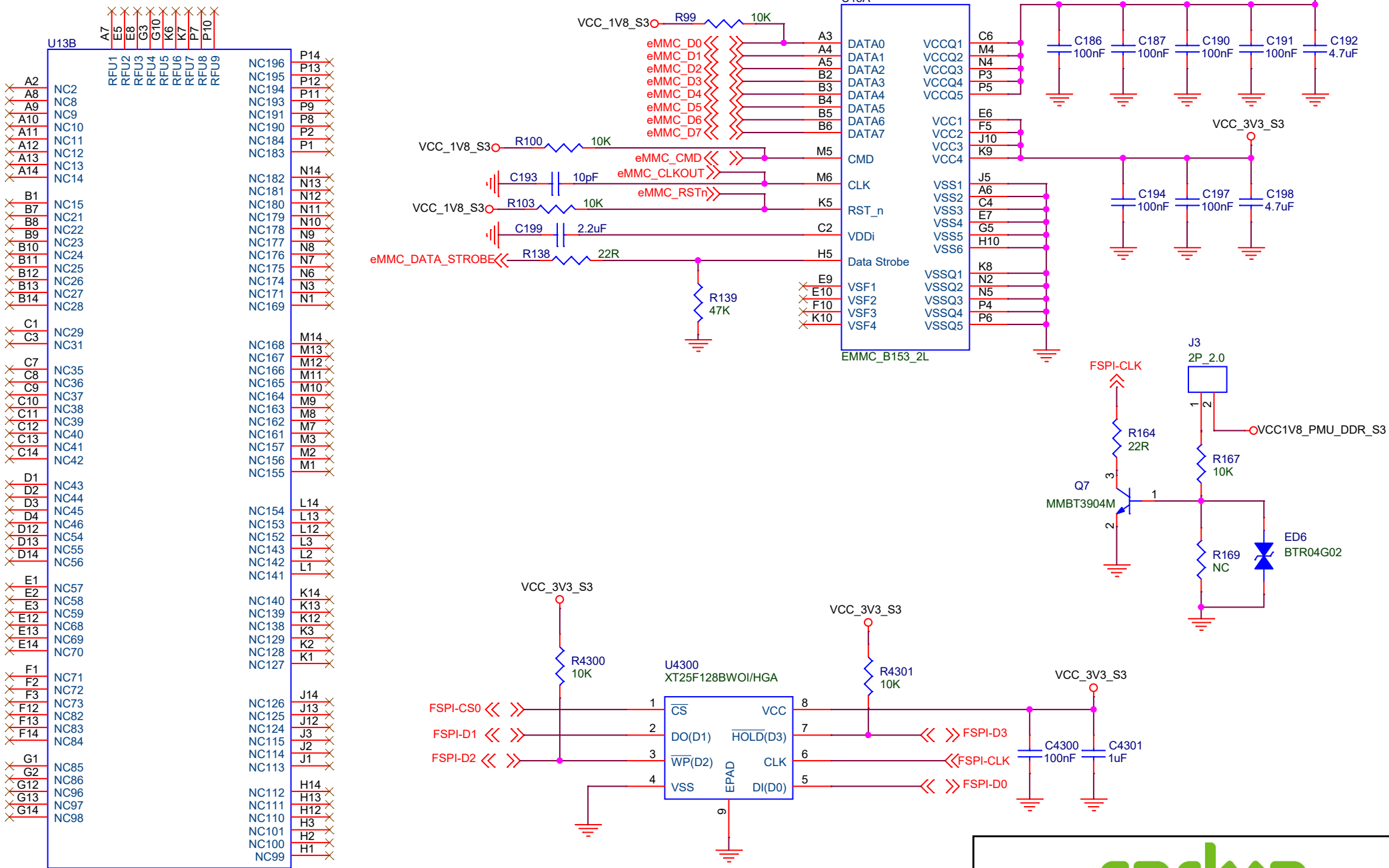

| | | | |
|---|---|---|---|
| Size A4 | Title: ROCK 5T | | REV 1.2 |
| | Page Name: VO-LCM_MIPI | | |
| Date: Thursday, January 09, 2025 | | Sheet 19 of 36 | |

# CODEC ES8316



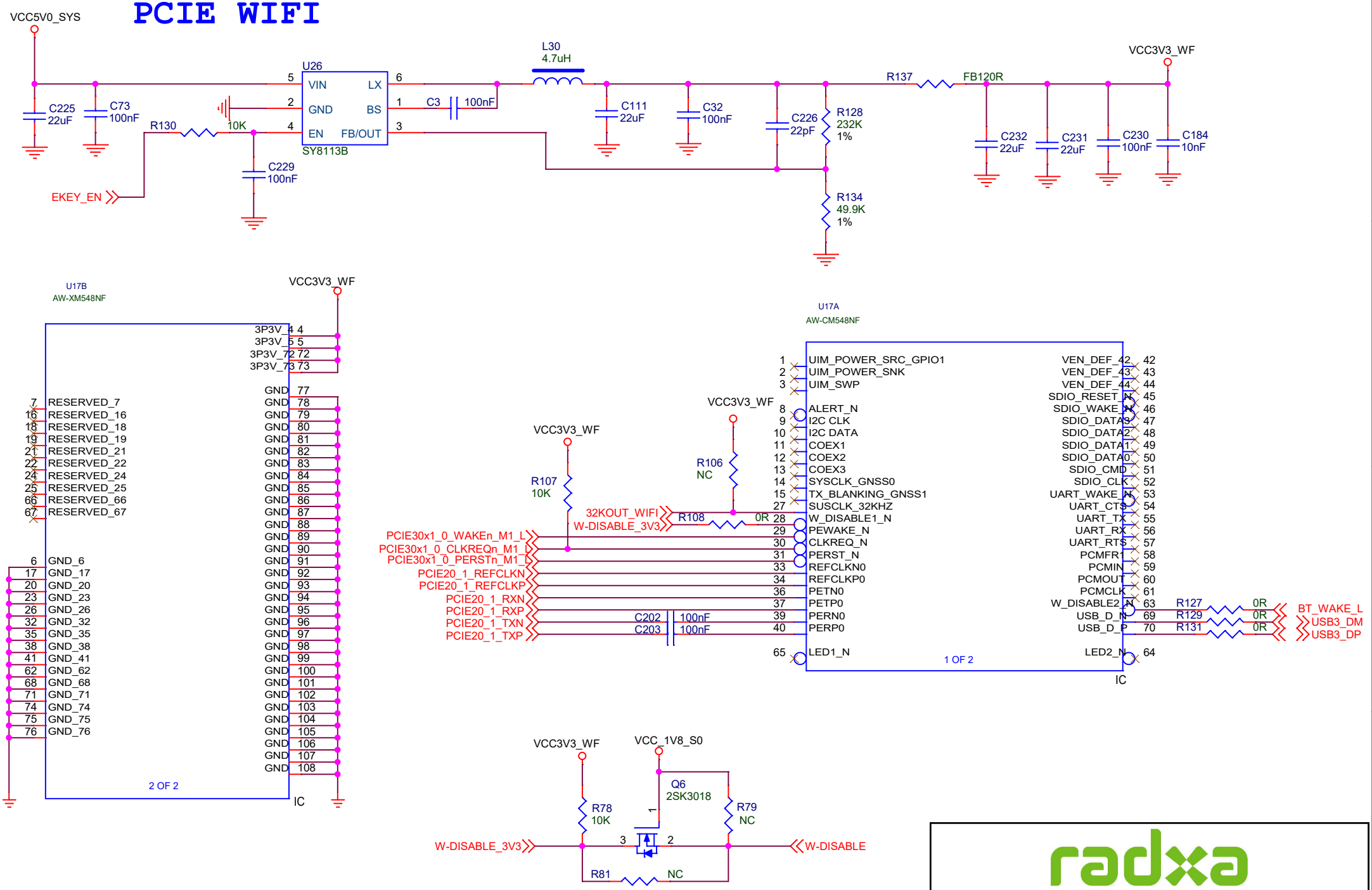

| | | | |
|---|---|---|---|
| Size A4 | Title: ROCK 5T | | REV 1.2 |
| | Page Name: Audio Codec | | |
| Date: Thursday, January 09, 2025 | | Sheet 20 of 36 | |

# eMMC FLASH



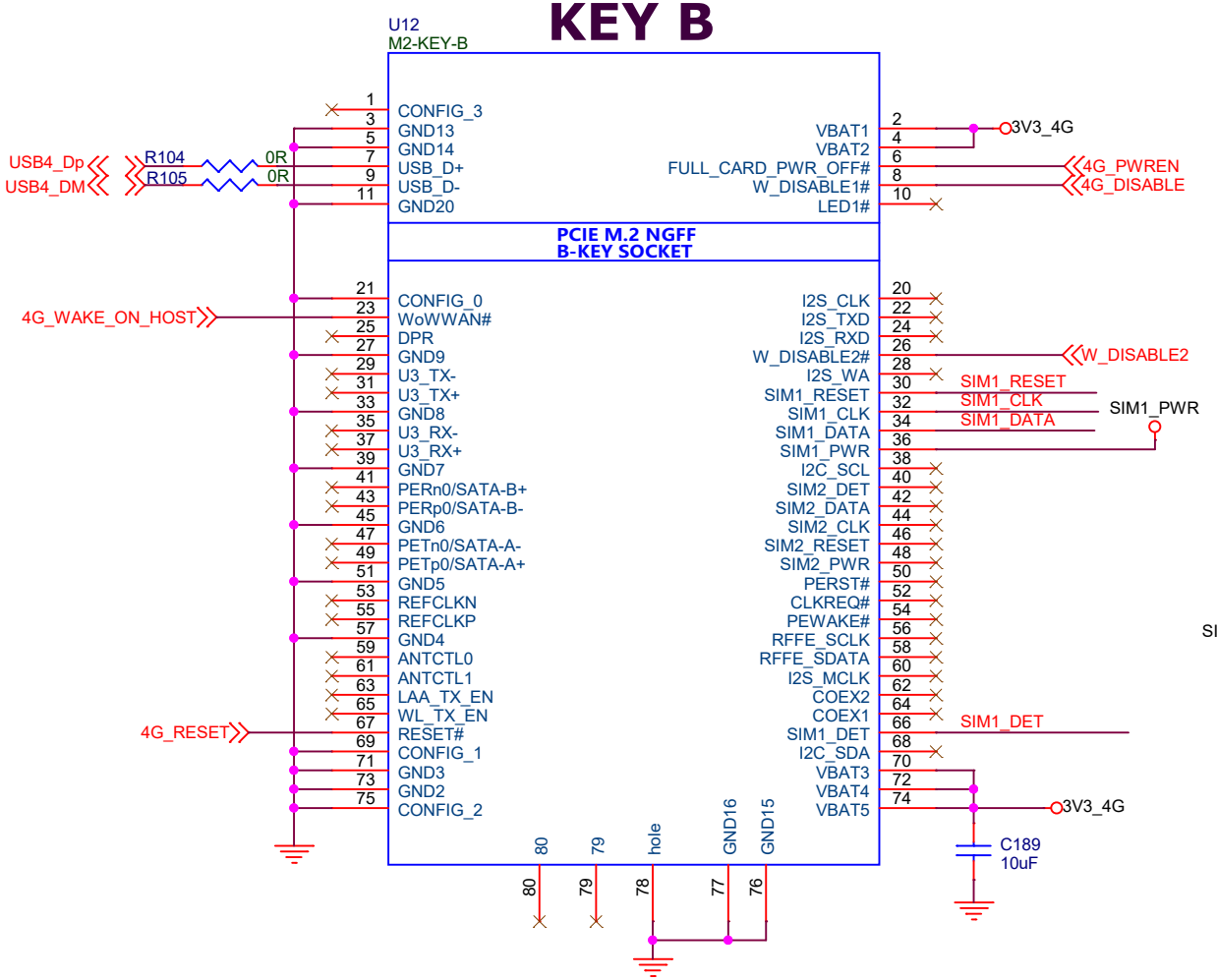

| | | | |
|---|---|---|---|
| Size A4 | Title: ROCK 5T | | REV 1.2 |
| | Page Name: SPI/eMMC Flash | | |
| Date: Thursday, January 09, 2025 | | Sheet 21 of 36 | |

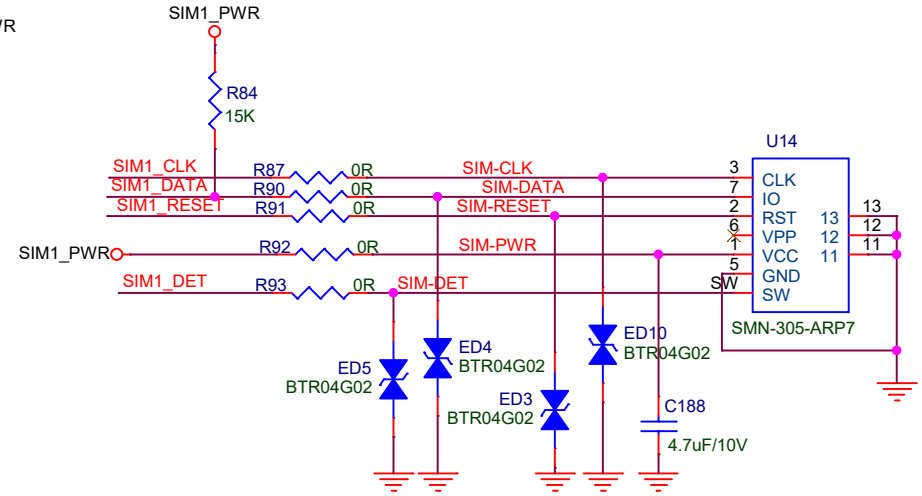
# PCIE WIFI

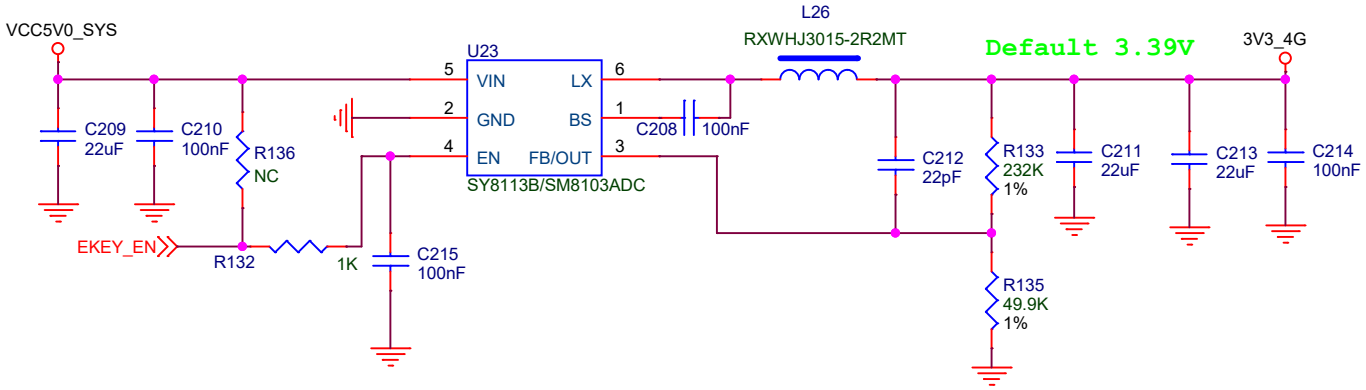



| | | | |
|---|---|---|---|
| Size A4 | Title: ROCK 5T | | REV 1.2 |
| | Page Name: PCIE_WIFI | | |
| Date: Thursday, January 09, 2025 | Sheet 22 of 36 | | |

# KEY B

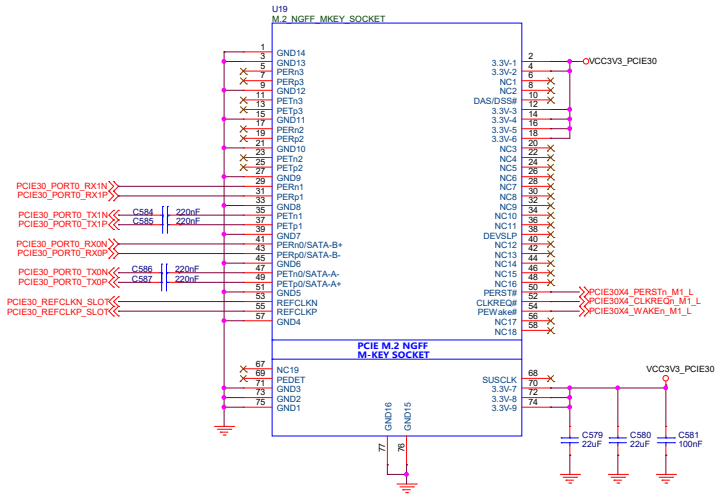


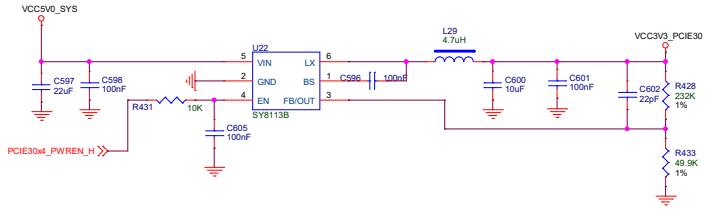



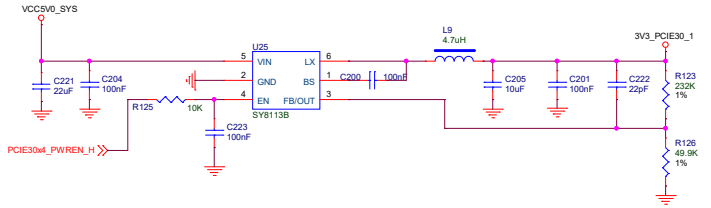




| | | | |
|---|---|---|---|
| Size A4 | Title: | ROCK 5T | REV 1.2 |
| | Page Name: | M.2 B key_4G | |
| Date: | Thursday, January 09, 2025 | Sheet 23 of 36 | |

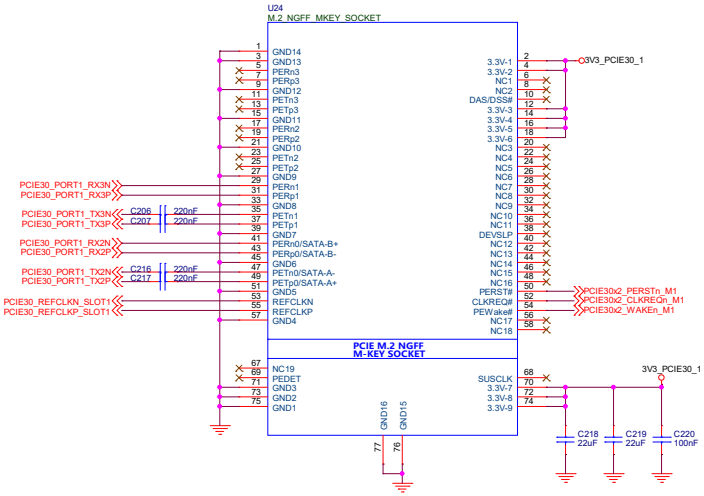
# PCIe3.0 x 2 Lane_1

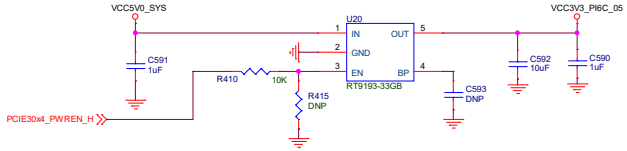


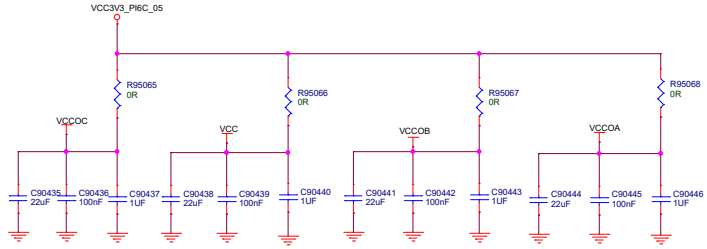



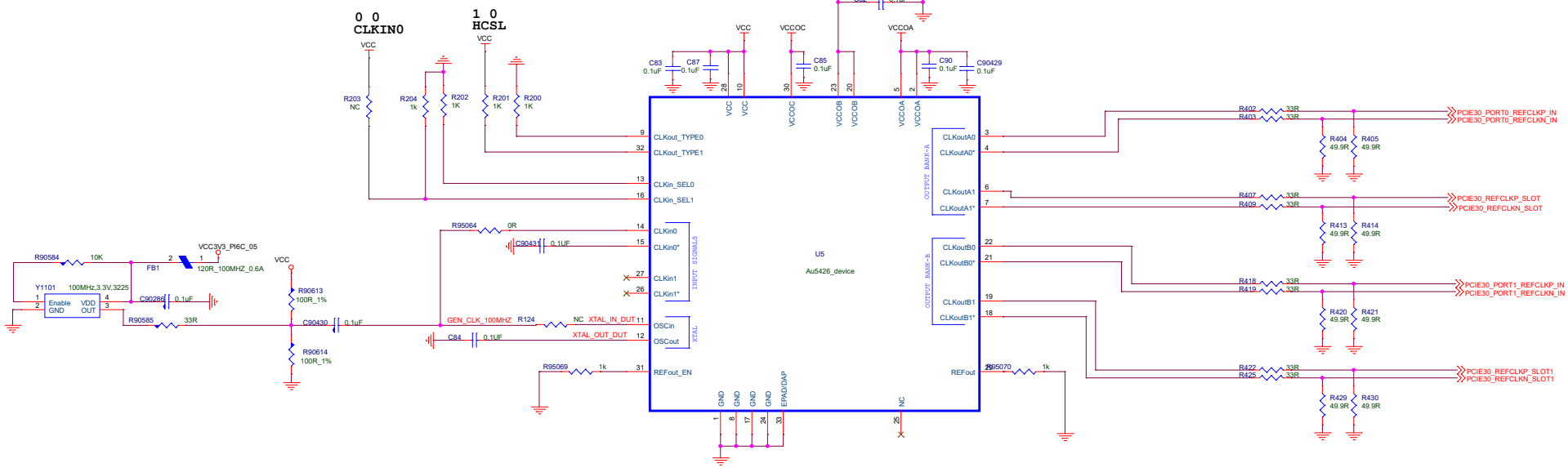




# PCIe3.0 x 2 Lane_2



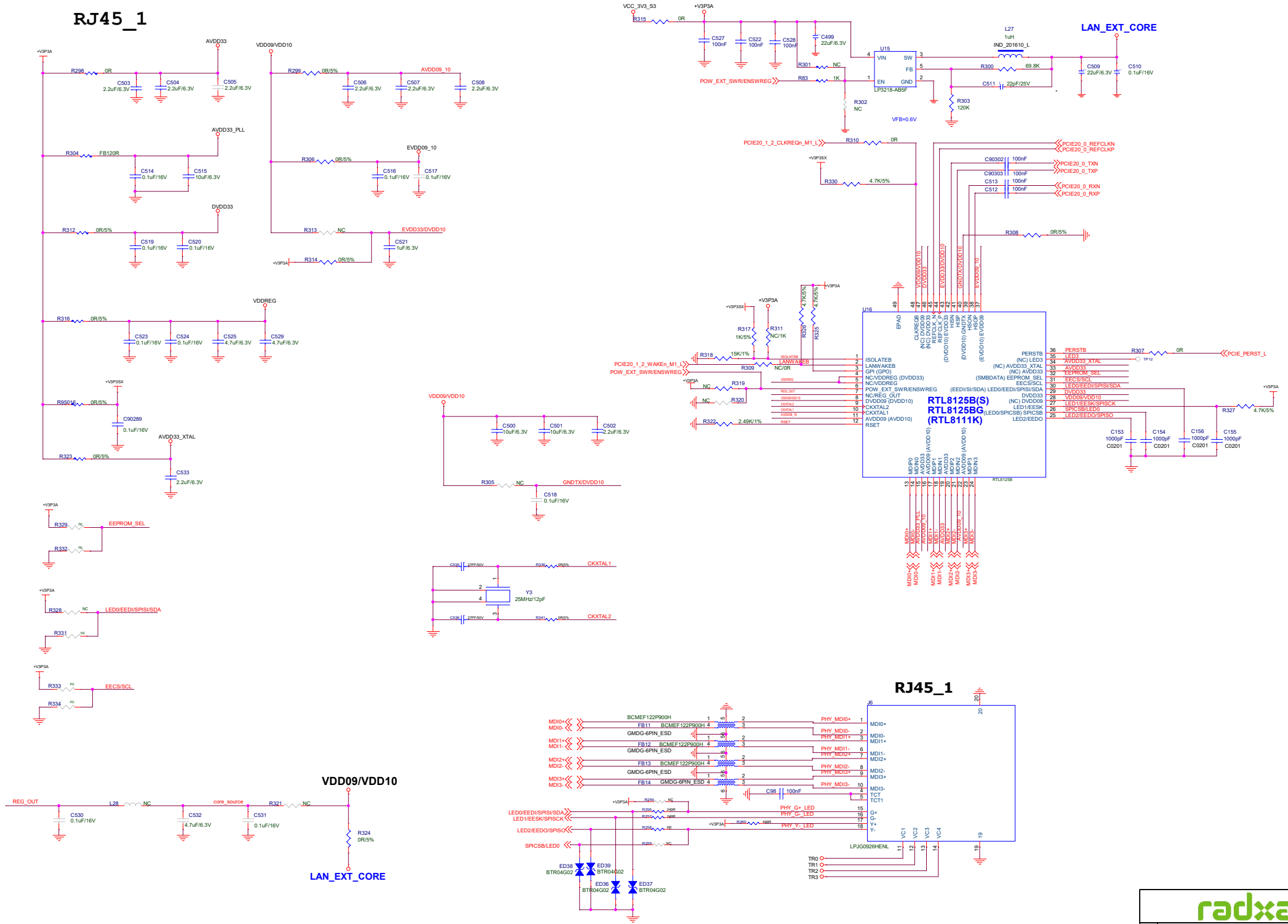




| | | | |
|---|---|---|---|
| Size | Title: | ROCK 5T | REV |
| A2 | Page Name: | PCIE3.0 M key X2 | 1.2 |
| Date: | Thursday, January 09, 2025 | Sheet 24 of 36 | |

# RJ45_1















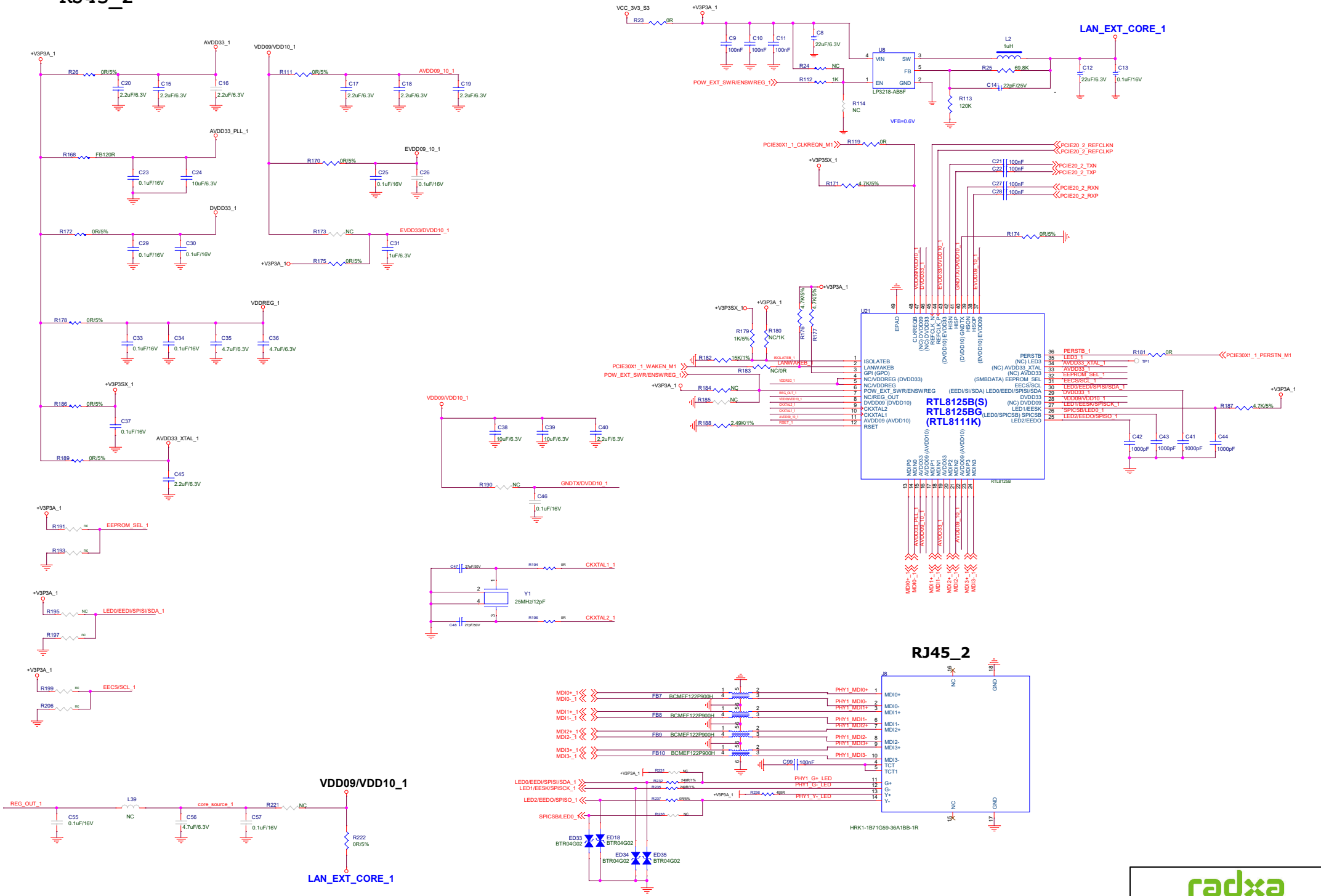






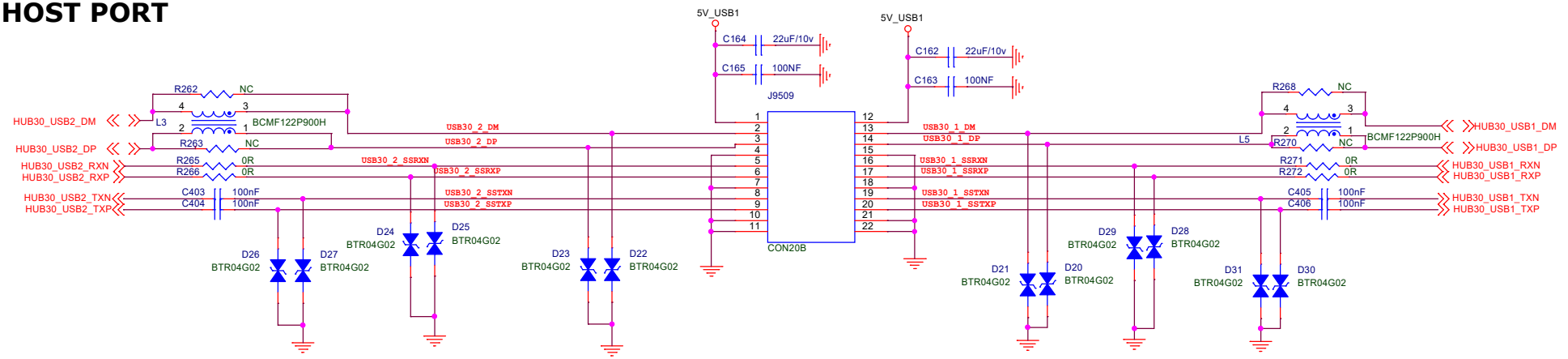

| | | | |
|---|---|---|---|
| Size | Title: ROCK 5T | | REV |
| A2 | Page Name: PCIE TO RJ45_1 | | 1.2 |
| Date: Thursday, January 09, 2025 | Sheet 25 of 36 | | |

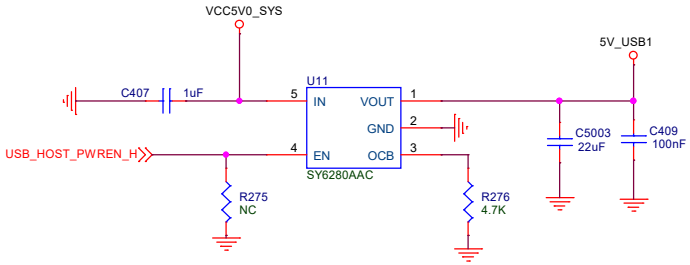
# RJ45_2

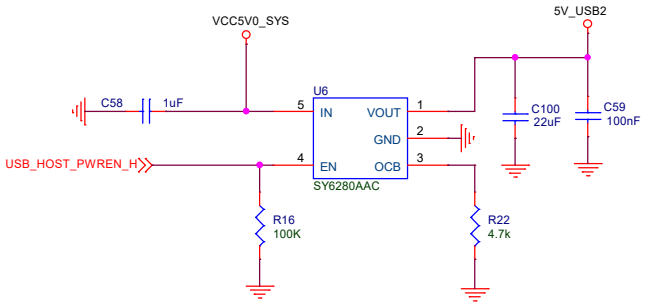


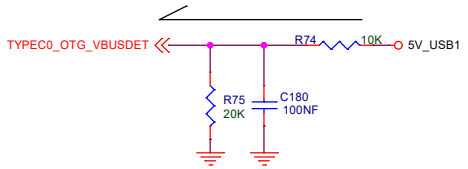

| | | |
|---|---|---|
| Size: A2 | Title: ROCK 5T<br>Page Name: PCIE TO RJ45_2 | REV: 1.2 |
| Date: Thursday, January 09, 2025 | Sheet 26 of 36 | |

# USB3.0 HOST PORT

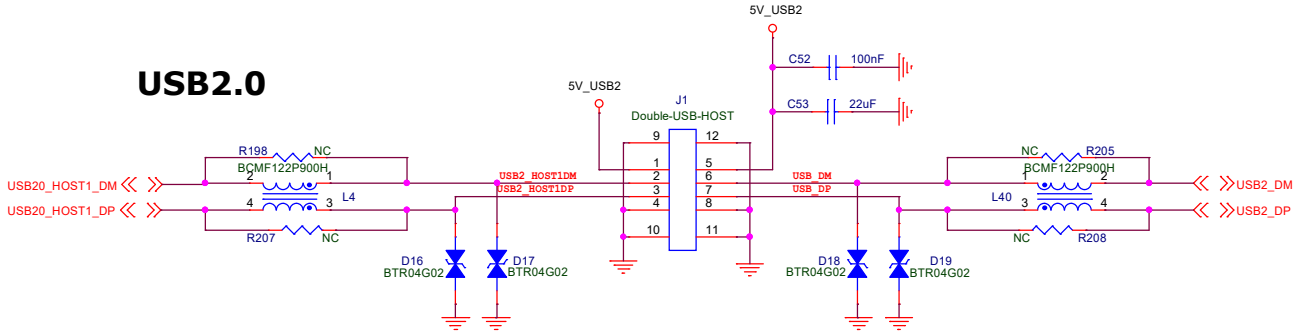


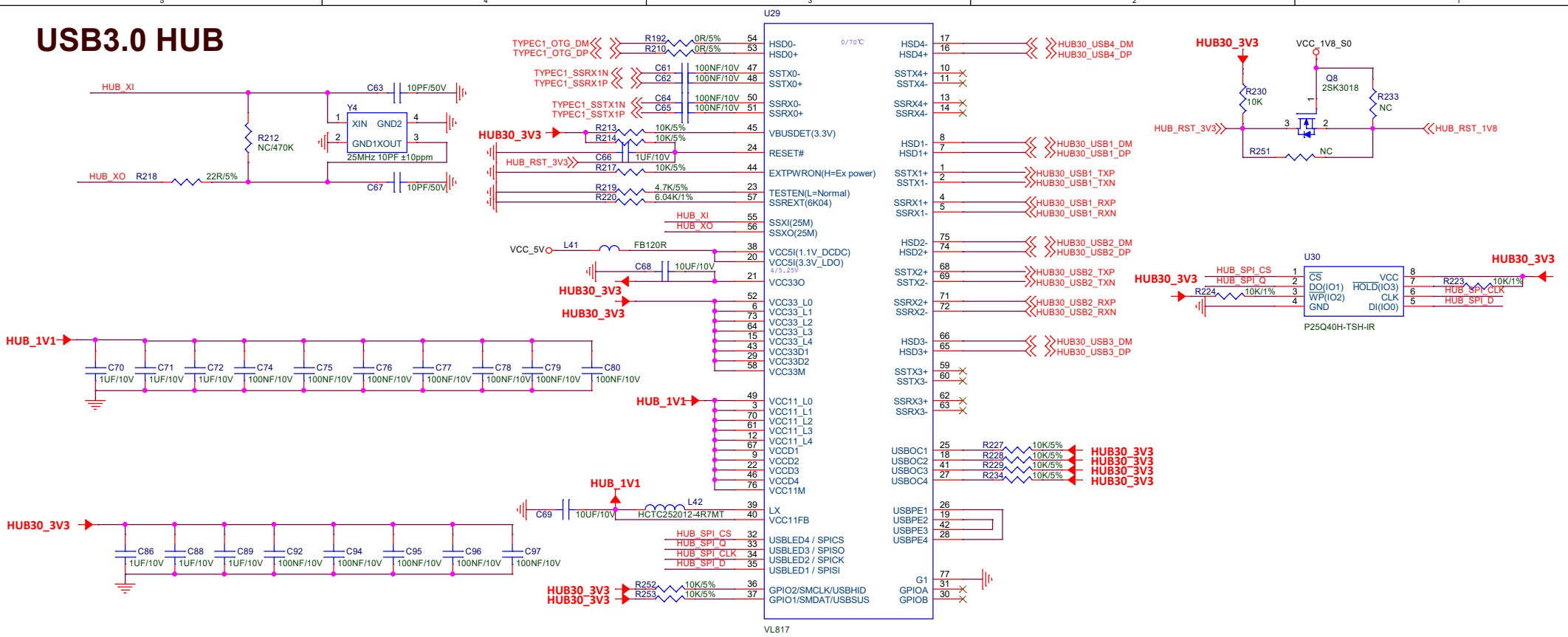




# USB2.0

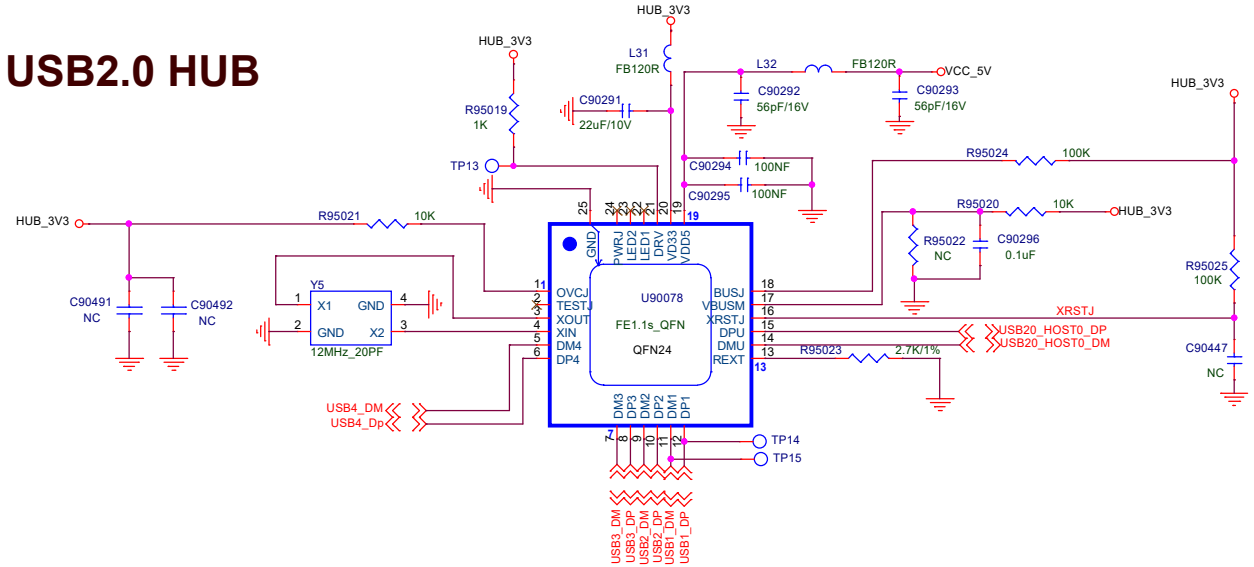



| | | | |
|---|---|---|---|
| Size | Title: | ROCK 5T | REV |
| A3 | Page Name: | USB30/20 Port | 1.2 |
| Date: | Thursday, January 09, 2025 | Sheet 27 of 36 | |

# USB3.0 HUB

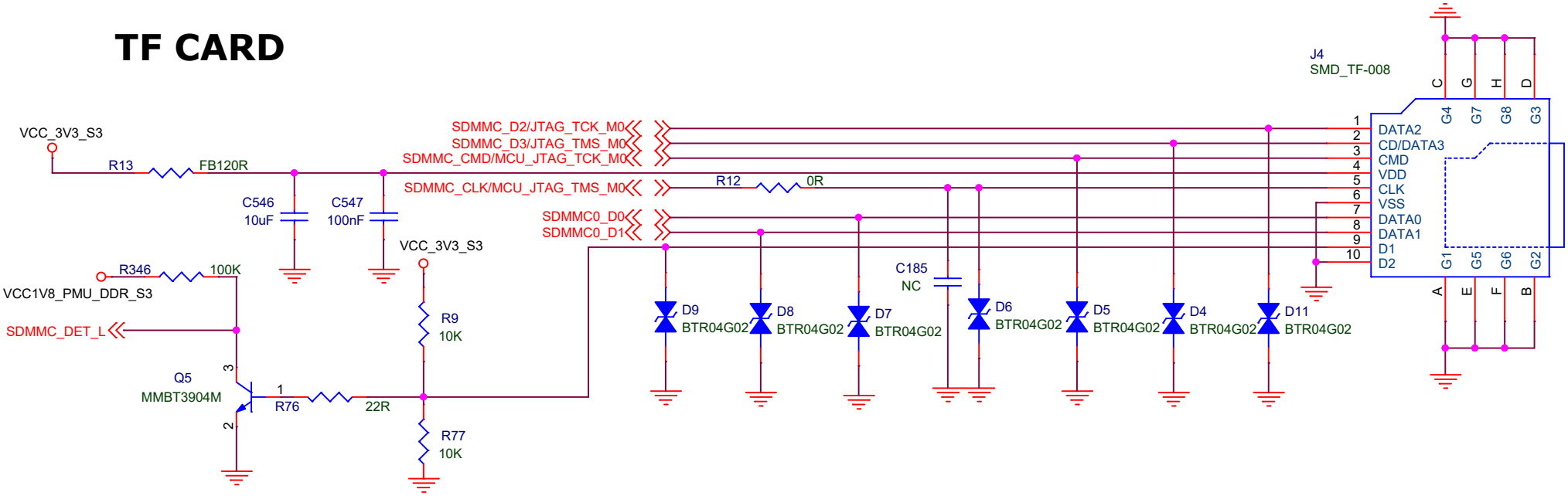



# USB2.0 HUB

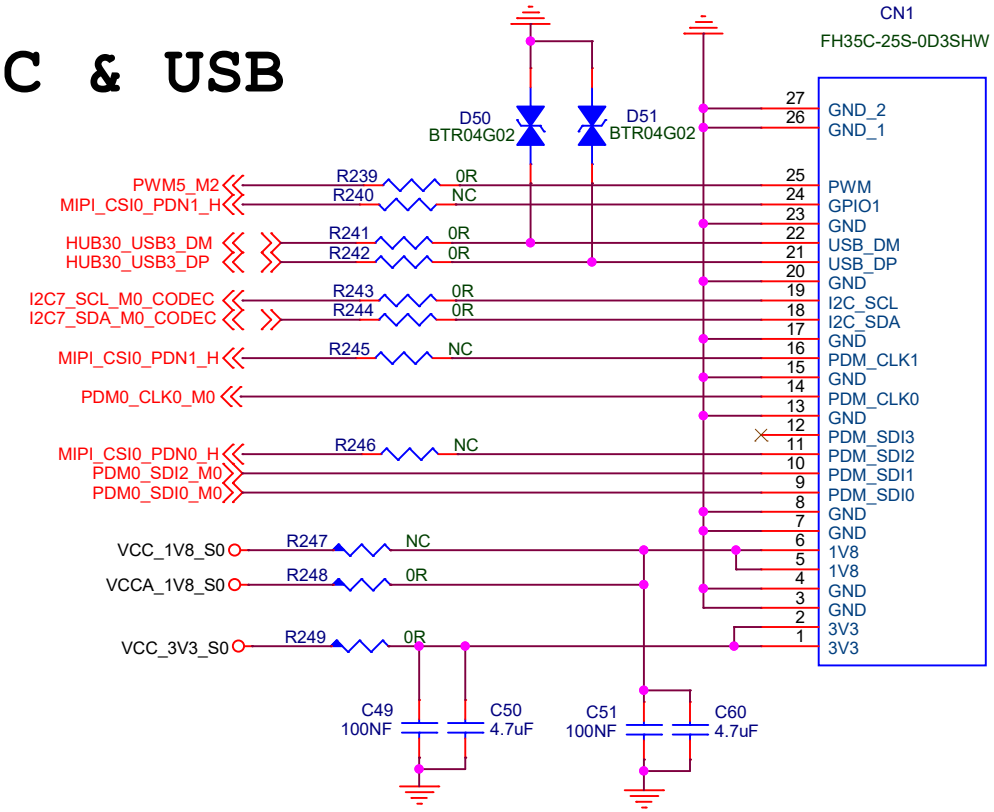



| | | | |
|---|---|---|---|
| Size | Title: ROCK 5T | | REV |
| A3 | Page Name: USB20/30 HUB | | 1.2 |
| Date: Thursday, January 09, 2025 | Sheet 28 of 36 | | |

# TF CARD

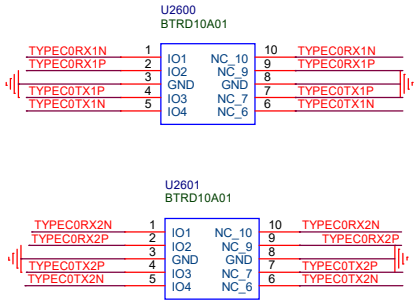


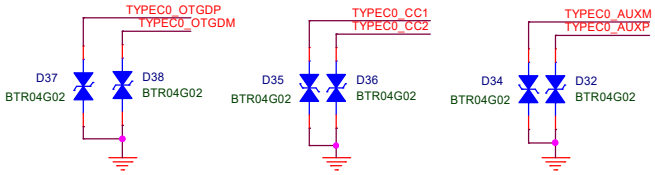

# PDM_DMIC & USB

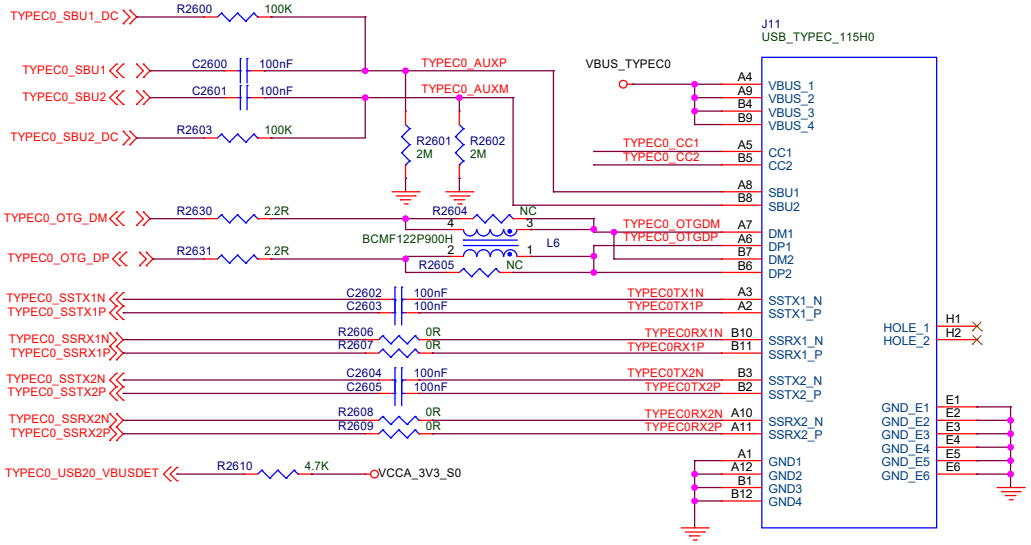


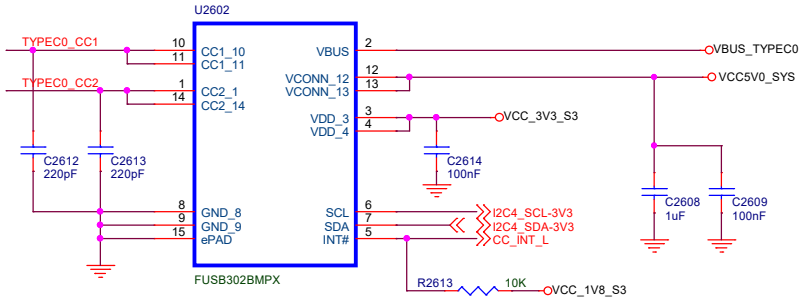

| Size | Title: | ROCK 5T | REV |
|---|---|---|---|
| A4 | Page Name: | Flash-TF Card/DMIC | 1.2 |
| Date: | Thursday, January 09, 2025 | Sheet 29 of 36 | |

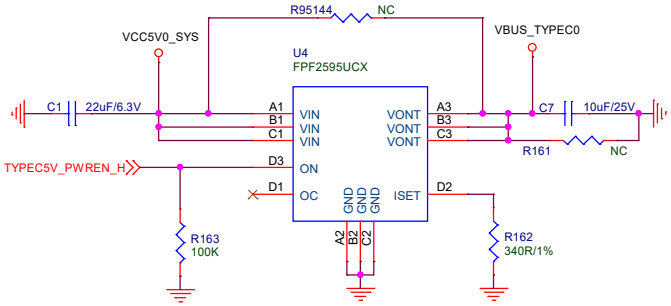
# Type-C PORT



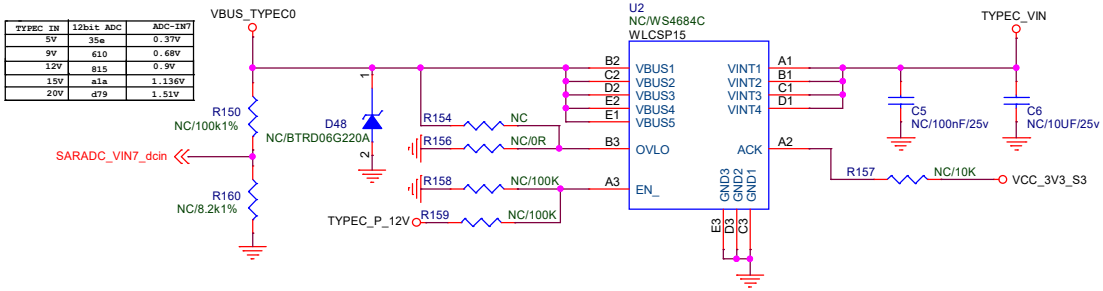



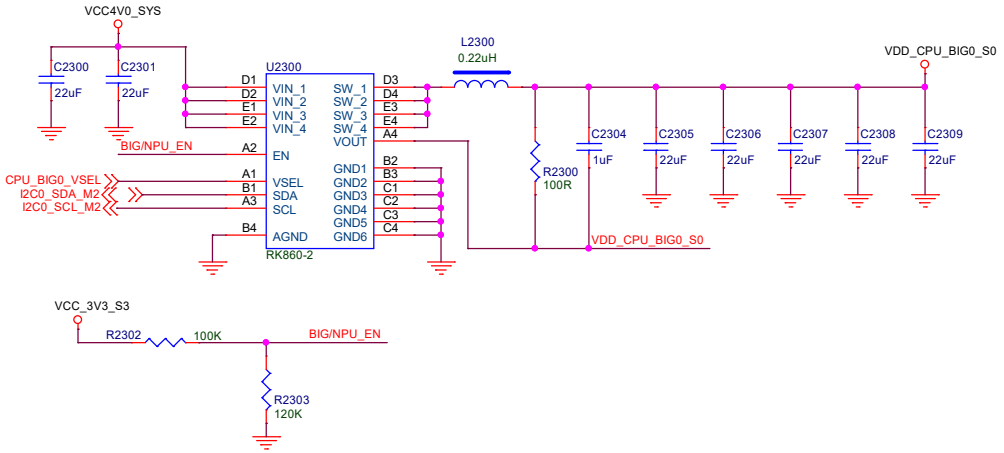

| TYPEC IN | 12bit ADC | ADC-IN7 |
|---|---|---|
| 5V | 35a | 0.37V |
| 9V | 610 | 0.68V |
| 12V | 815 | 0.9V |
| 15V | a1a | 1.136V |
| 20V | d79 | 1.51V |



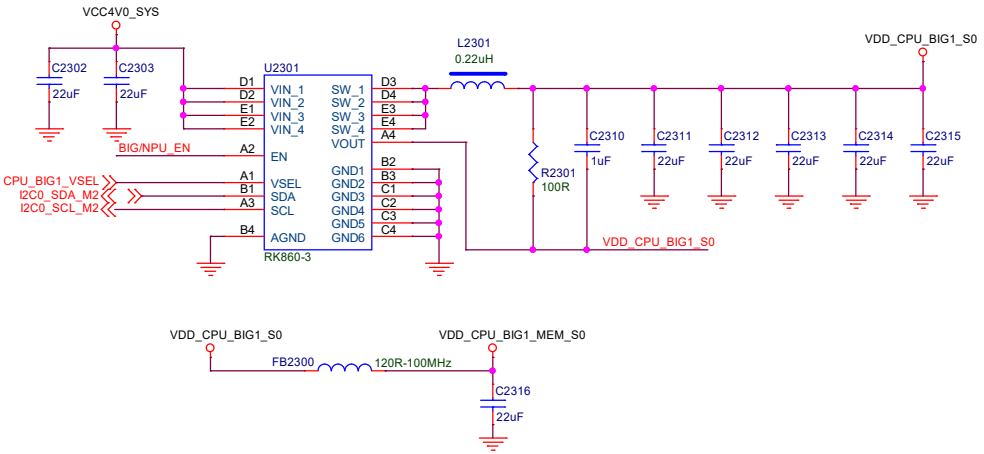

**TYPEC_POW1>VBUS_TYPEC0**

| | | | |
|---|---|---|---|
| radxa | Size: A3 | Title: ROCK 5T | REV: 1.2 |
| | | Page Name: Type-C Port | |
| | Date: Thursday, January 09, 2025 | Sheet 30 of 36 | |

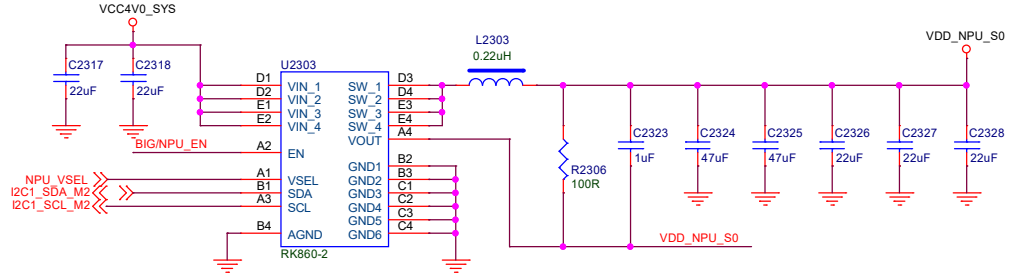
# VDD_CPU_BIG0



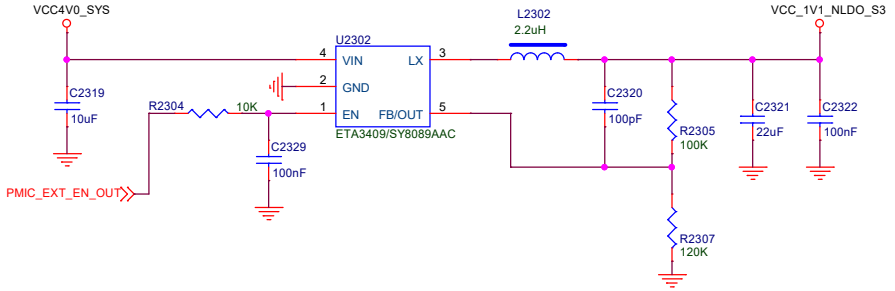

# VDD_CPU_BIG1

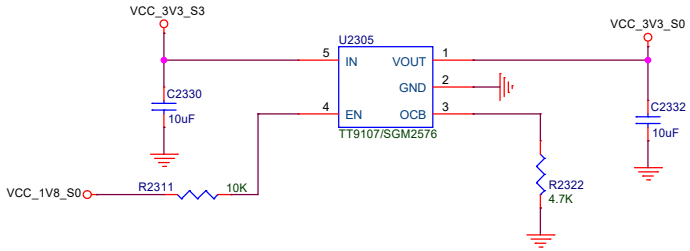


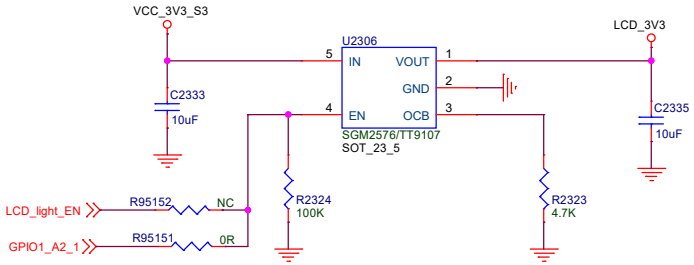

# VDD_NPU




# VCC_1V1_NLDO_S3

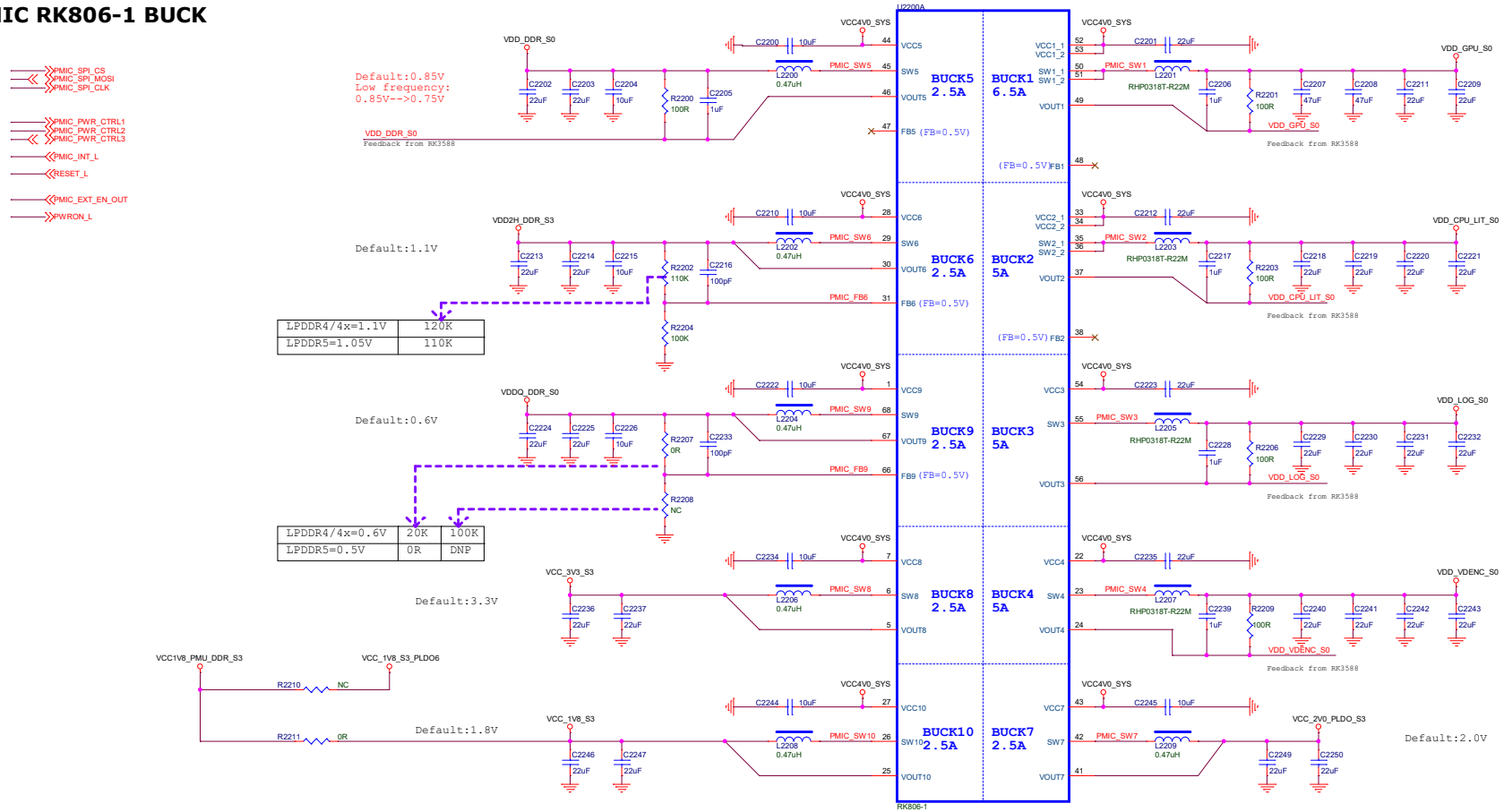


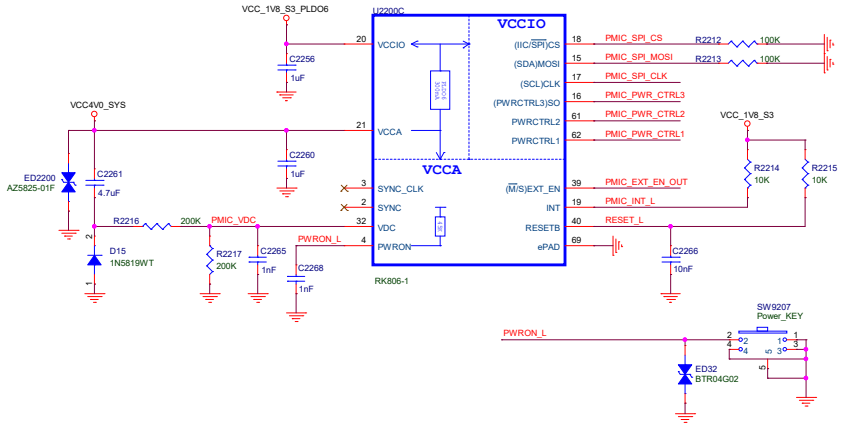




| | | | |
|---|---|---|---|
| Size A3 | Title: ROCK 5T | | REV 1.2 |
| | Page Name: Power_Ext Discrete | | |
| Date: Thursday, January 09, 2025 | Sheet 31 of 36 | | |

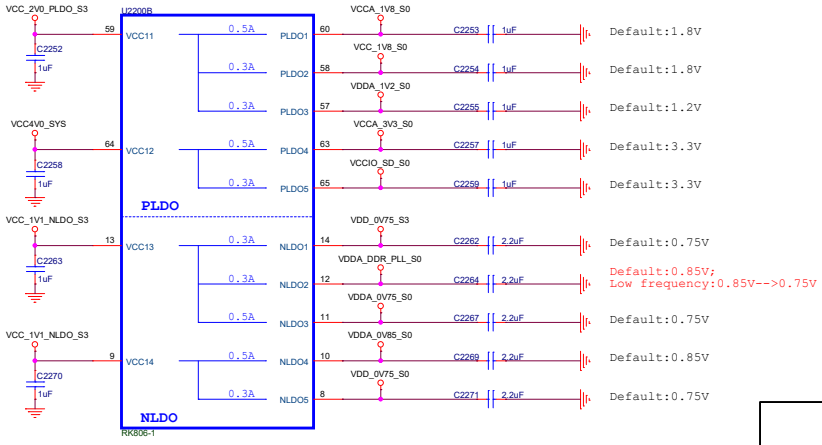
# PMIC RK806-1 BUCK




# PMIC RK806-1 Managerment

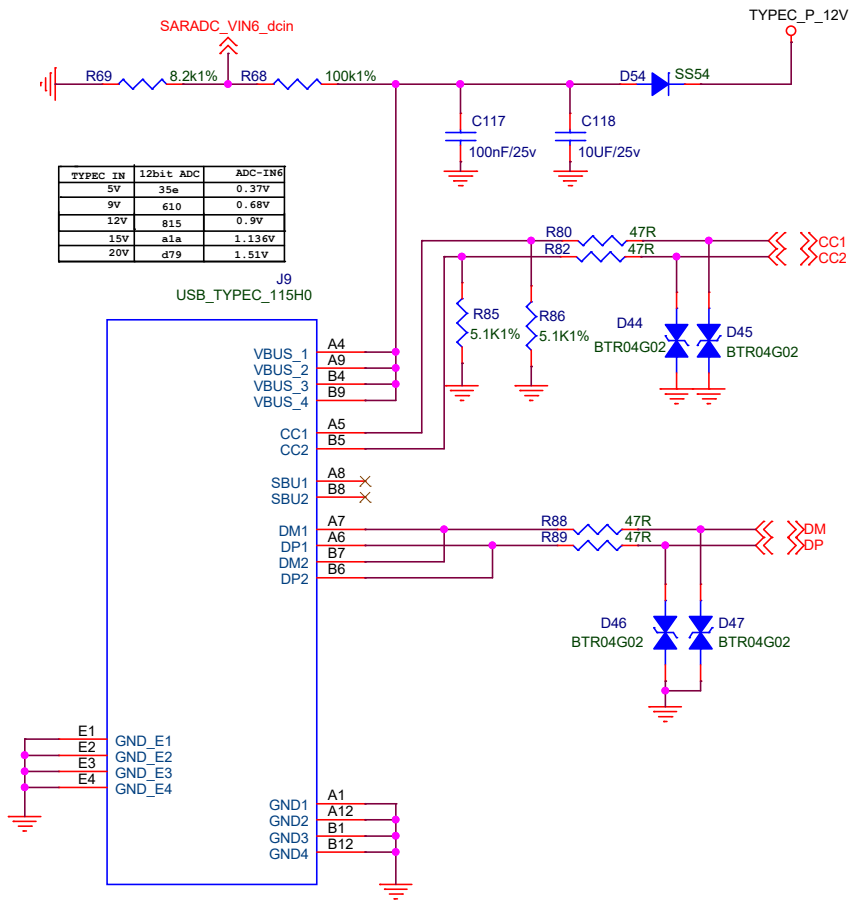



# PMIC RK806-1 LDO

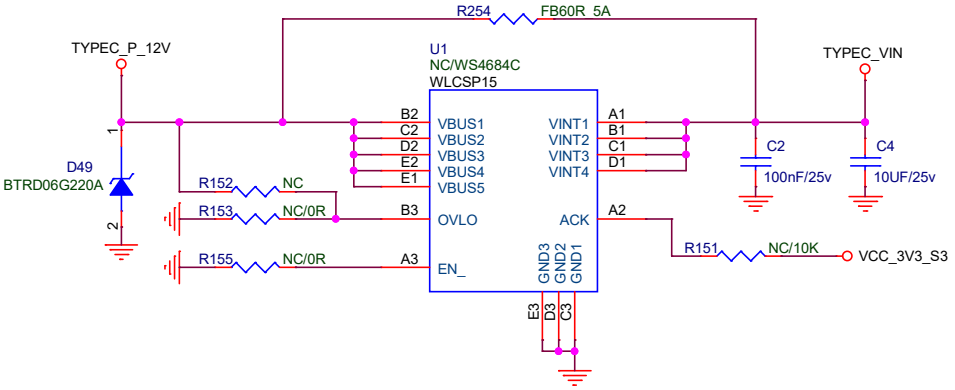


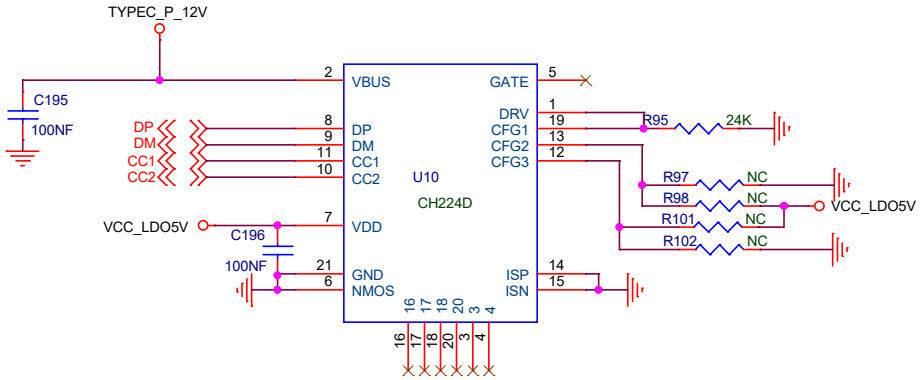

| | | |
|---|---|---|
| radxa | | |
| Size: A2 | Title: ROCK 5T | REV: 1.2 |
| | Page Name: Power-PMIC_RK806-1 | |
| Date: Thursday, January 09, 2025 | Sheet 32 of 36 | |

# 12V/3A DCIN



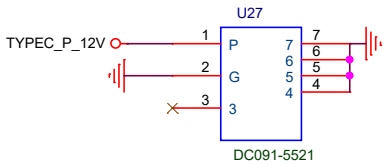

| TYPEC IN | 12bit ADC | ADC-IN6 |
|---|---|---|
| 5V | 35e | 0.37V |
| 9V | 610 | 0.68V |
| 12V | 815 | 0.9V |
| 15V | a1a | 1.136V |
| 20V | d79 | 1.51V |



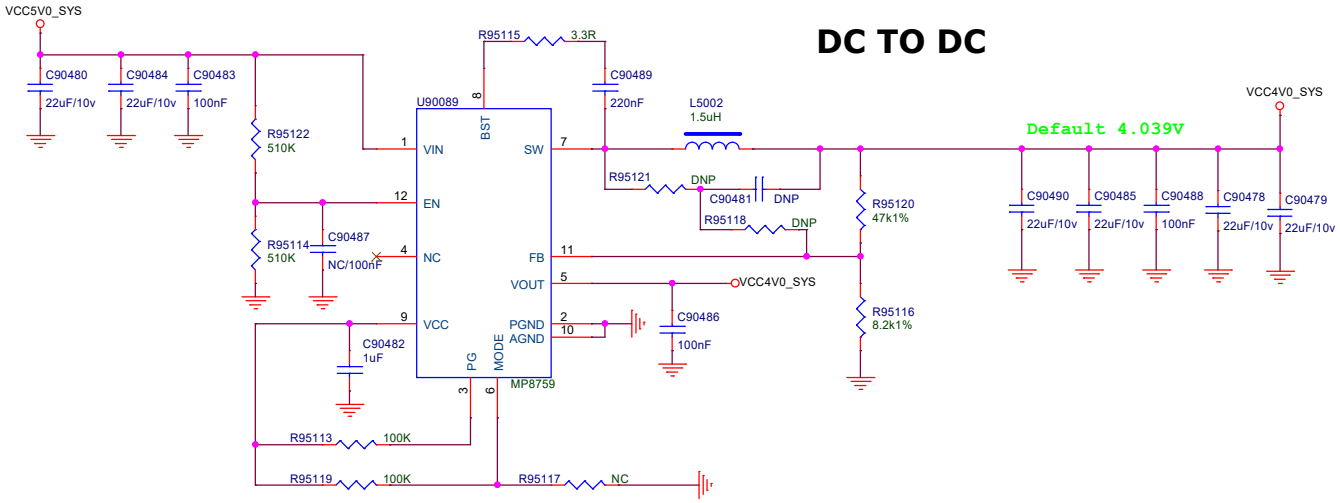


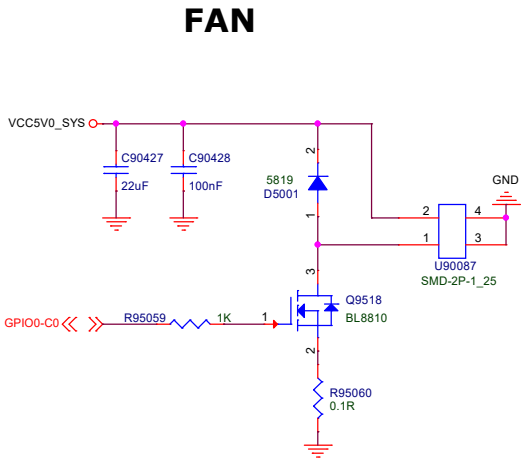


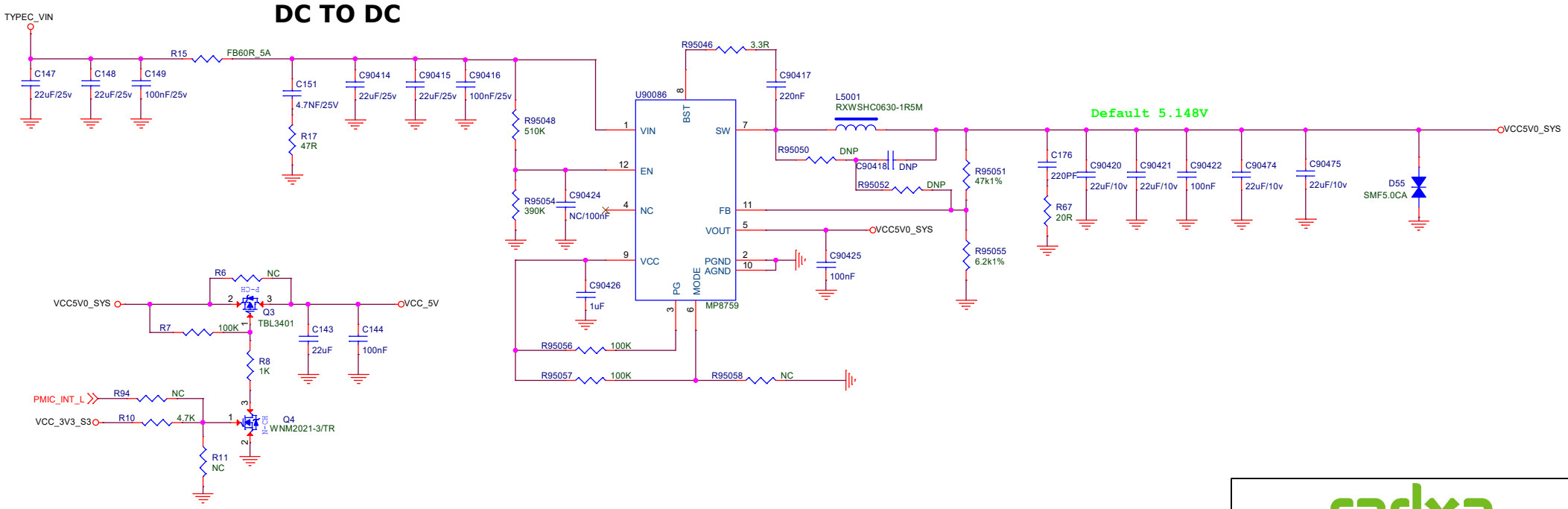



radxa

| Size | Title: | REV |
|---|---|---|
| Custom | ROCK 5T | |
| | Page Name: Power TYPEC IN | 1.2 |
| Date: Thursday, January 09, 2025 | Sheet 33 of 36 | |



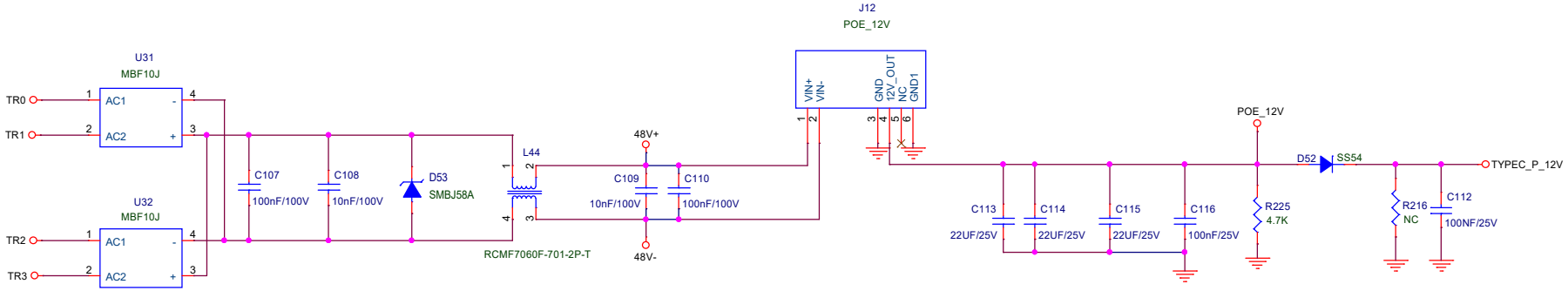




radxa

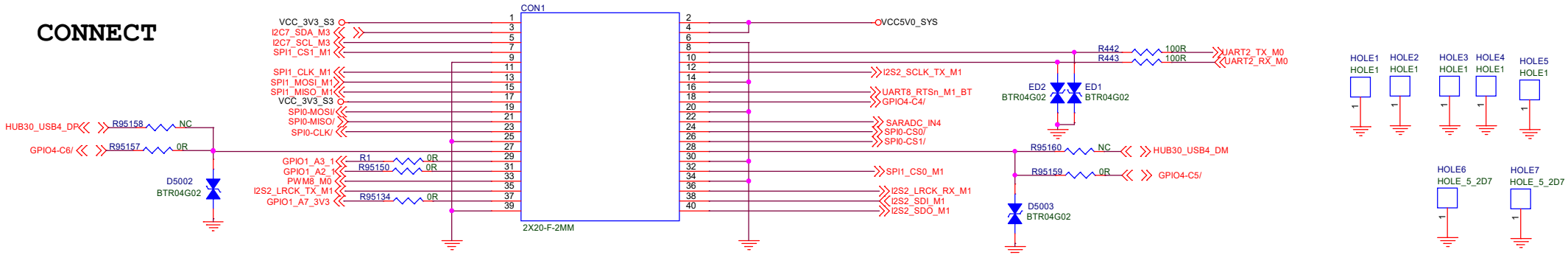
| Size | Title: | REV |
|---|---|---|
| A3 | ROCK 5T | |
| | Page Name: Power_DC 5V/FAN | 1.2 |
| Date: Thursday, January 09, 2025 | Sheet 34 of 36 | |

# POE POWER

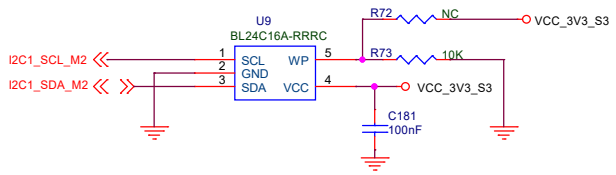



| Size | Title: | REV |
|---|---|---|
| A3 | ROCK 5T | |
| | Page Name: POE IN | 1.2 |
| Date: Thursday, January 09, 2025 | Sheet 35 of 36 | |

# CONNECT



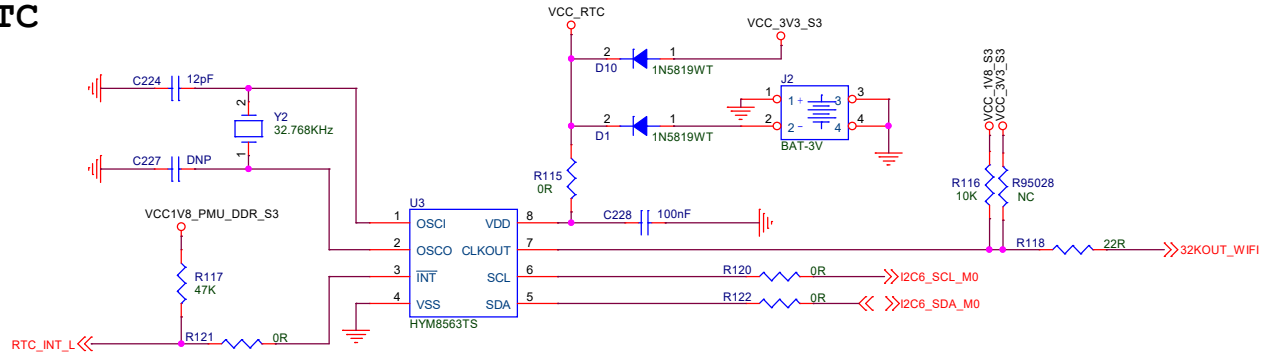

# EEPROM

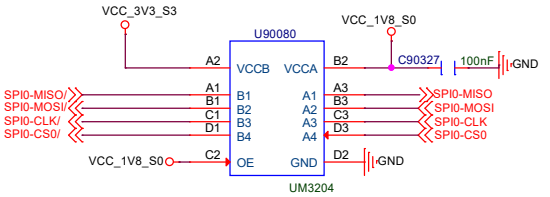


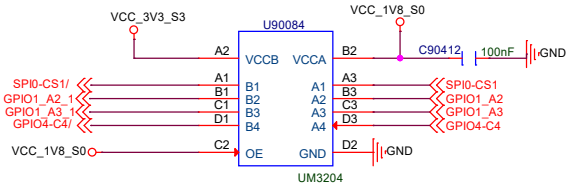




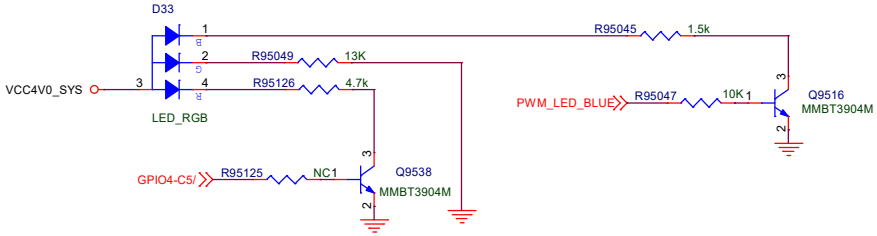
# RTC

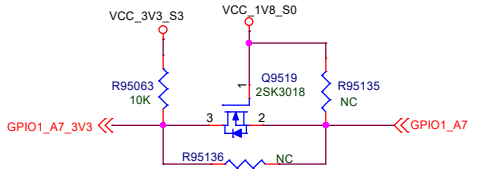



# LED




| | | |
|---|---|---|
| Size A3 | Title: ROCK 5T | REV 1.2 |
| | Page Name: 40pin/LED/RTC/EEPROM | |
| Date: Thursday, January 09, 2025 | Sheet 36 of 36 | |